

## Chapter 60

### Touch Sensor (TOUCH)

ESP32-P4 has 14 capacitive touch sensor units, each of which can be connected to an external touch panel via a touch pin (GPIO pin) to constitute a touch sensor system. The ESP32-P4 touch sensor system is mainly used in human-computer interactions to detect finger touch or proximity. It also features moisture resistance and waterproof design.

#### 60.1 Terminology

To better illustrate the functions of capacitive touch sensors, the following terms are used in this section.

|                            |                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Touch Sensor</b>        | Touch-related internal sensing circuitry.                                                        |
| <b>Touch Pin</b>           | Pins with touch sensing feature.                                                                 |
| <b>Touch Panel</b>         | The external device connected to touch sensor via channel (touch pin) to detect finger touch.    |
| <b>Touch Sensor System</b> | Capacitive touch sensing system, consisting of touch sensor, touch pin, traces, and touch panel. |

**Note:**

In subsequent description, “touch panel is sampled/scanned/measured” indicates the same action to touch pin, i.e. “touch pin is sampled/scanned/measured”.

#### 60.2 Feature List

The ESP32-P4 touch sensor has the following features:

- Detection of 14 capacitive touch pins
- Sampling triggered by software or dedicated hardware timer
- Two sampling methods:
  - Pulses from the touch pins used as clock signals to count the sampling period
  - Pulses from the touch pins used as digital signals; sample the rising edge of the digital signal with the system clock to count the sampling period
- Scan mode, supporting sequential sampling of multiple touch pins by configuring the Touch FSM.
- Timeout mechanism to monitor channel abnormality
- Frequency hopping to increase the anti-interference of detection
- Proximity sensing mode with up to three configurable channels

- Configuration of individual touch sensors to operate normally in sleep mode
- Wake-up by touch sensor
- Moisture resistance
- Waterproof design

## 60.3 Architectural Overview

### 60.3.1 Touch Panel

A touch panel consists of the following components:

- Electrode: Changes in capacitance when touched by a finger.
- Substrate: The base material on which the protective cover, electrode, and the electrode's connector to the channel are built.
- Protective cover: To physically separate the other components from the external environment.

When developing applications using the touch sensing feature, users should decide on the placement, the material, or the arrangement of the touch panels during mechanical design. For more information about the design guidelines, please refer to [Touch Sensor Application Note](#). Touch panel can be connected to ESP32-P4 touch sensor via touch pin (channel). See Figure 60.3-1.

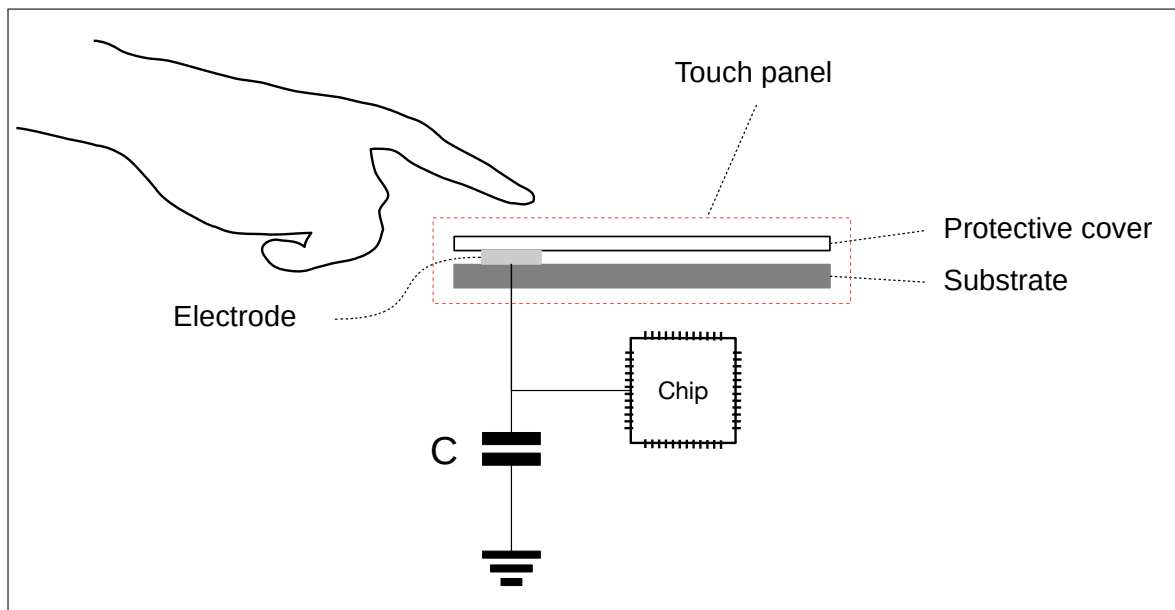


Figure 60.3-1. Touch Sensor System

When users touch the protective cover, the capacitance of the electrode increases. If the electrode is connected to one of the ESP32-P4 touch sensors via a channel, the touch sensor will be able to detect the electrode's change in capacitance. If the change of capacitance exceeds the configurable threshold, the touch sensor will trigger an interrupt.

### 60.3.2 Capacitive Touch Pin

ESP32-P4 provides 14 capacitive touch pins labeled as T1 ~ T14, each connected to the chip's pin to monitor finger touch from the external environment. The mapping between the touch sensors and the chip pins is shown in Table 60.3-1.

Table 60.3-1. ESP32-P4 Capacitive Touch Pins

| Capacitive Touch Pin | Connected GPIO Pin |
|----------------------|--------------------|
| T1                   | GPIO2              |
| T2                   | GPIO3              |
| T3                   | GPIO4              |
| T4                   | GPIO5              |
| T5                   | GPIO6              |
| T6                   | GPIO7              |
| T7                   | GPIO8              |
| T8                   | GPIO9              |
| T9                   | GPIO10             |
| T10                  | GPIO11             |
| T11                  | GPIO12             |
| T12                  | GPIO13             |
| T13                  | GPIO14             |
| T14                  | GPIO15             |

### 60.3.3 Touch Sensor

The touch sensor charges and discharges the touch pin with a fixed current source, and each charge and discharge generates a pulse signal, the TOUCH\_OUT signal. The touch sensor detects the capacitance change of the touch pin. If the touch pin is touched or approached by a finger, the touch pin's capacitance increases and the charging and discharging time is extended. By measuring the time required for charging and discharging the touch pin a fixed number of times, it can be inferred whether the touch pin has been touched or not.

Figure 60.3-2 illustrates the internal structure of the touch sensors. Each of the touch sensors has a set of input and output signals, some of which are connected to the Touch FSM (see Section 60.4.1). Table 60.3-2 shows the output signals of the touch sensor and their functions.

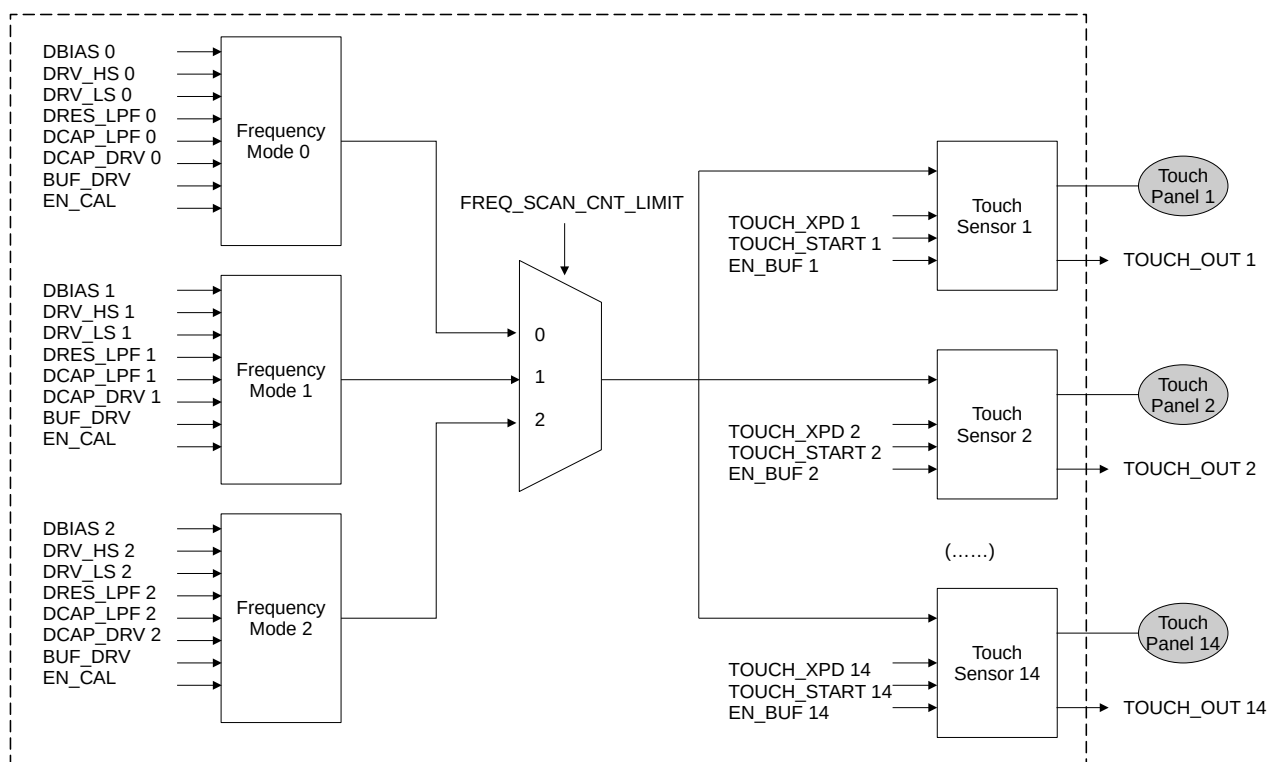


Figure 60.3-2. Touch Sensor Internal Structure

Table 60.3-2. Capacitive Touch Sensor Output Signal

| Signal      | Source    | Function                                                                                                                                                                                     |
|-------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TOUCH_XPD   | Touch FSM | Controls the power-up and power-down of the touch sensor                                                                                                                                     |
| TOUCH_START | Touch FSM | Controls the touch sensor to initiate a measurement. The measurement ends when the Touch FSM pulls down the TOUCH_START signal                                                               |
| DBIAS       | Touch FSM | The FSM selects different DBIAS based on the frequency mode to adjust the voltage of the touch sensor's internal LDO, which affects the charging and discharging speed of the touch sensor   |
| DRV_HS      | Touch FSM | The FSM selects different DRV_HS based on the frequency mode to adjust the high-speed driving capability of the touch sensor                                                                 |
| DRV_LS      | Touch FSM | The FSM selects different DRV_LS based on the frequency mode to adjust the low-speed driving capability of the touch sensor                                                                  |
| DRES_LPF    | Touch FSM | The FSM selects different DRES_LPF based on the frequency mode to adjust the RC low-pass filter, which can be configured as 0 k $\Omega$ , 3 k $\Omega$ , 4.5 k $\Omega$ , or 7.5 k $\Omega$ |
| DCAP_LPF    | Touch FSM | The FSM selects different DCAP_LPF based on the frequency mode to adjust the RC low-pass filter with an adjustment range of 0-2.54 pF and a step of 20 fF                                    |
| DCAP_DRV    | Touch FSM | Used to adjust the internal capacitance connected to touch channels during measurement, usually set to 0                                                                                     |
| EN_CAL      | Touch FSM | Used to disconnect from the touch pins and to test the capacitance in the touch sensor (the touch panel needs to be disabled)                                                                |

| Signal    | Source       | Function                                                                                                                                |
|-----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| BUF_DRV   | Touch FSM    | Adjusts the driving capability of TOUCH_BUFF                                                                                            |
| EN_BUF    | Touch FSM    | Enables TOUCH_BUFF, a software-configurable touch panel, to mitigate the effects of water droplets on the sensor for moisture tolerance |
| TOUCH_OUT | Touch sensor | The Touch FSM counts the pulses in the TOUCH_OUT signal and measures the time it takes to generate N pulses                             |

## 60.4 Functional Description

### 60.4.1 Touch FSM

The Touch FSM performs a measurement by selecting a touch sensor and controlling the necessary signals to and from a touch sensor. Figure 60.4-1 shows the internal structure and operating mechanism of the Touch FSM. The Touch FSM is clocked by LP\_DYN\_FAST\_CLK. For more information on LP\_DYN\_FAST\_CLK, see Chapter 10 *Reset and Clock*.

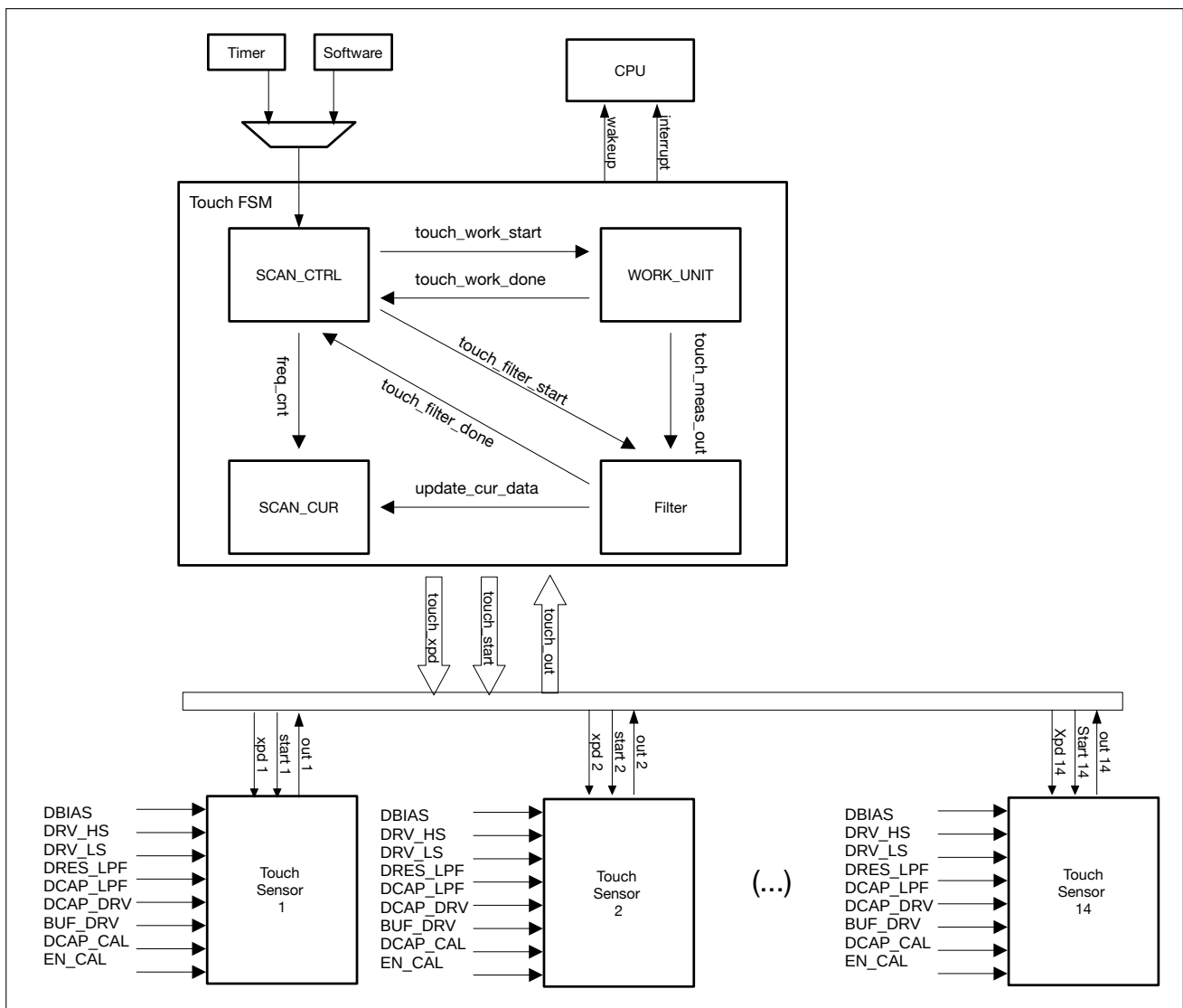


Figure 60.4-1. Touch FSM Structure and Mechanism

The following introduces the modules in the Touch FSM.

- **SCAN\_CTRL**: Select the touch pin to measure, set the parameters such as hopping frequency, and controls the start and end of measurement for each touch sensor in scan mode.
- **WORK\_UNIT**: Sample the selected touch sensors during a measurement.
- **Filter**: If enabled, each touch sensor can filter a series of measurements via an infinite impulse response (IIR) filter. The filtered value will be returned as the sampled value. See Section 60.4.3.1 for details.
- **SCAN\_CUR**: Cache the parameters of different touch sensors at different frequency modes during the measurement to be used for the next measurement.

## 60.4.2 Sampled Signal Preprocessing

The core mechanism of the touch sensor is to reflect whether the touch panel is touched or not based on the change of the output pulse wave, i.e., the TOUCH\_OUT signal, through detecting the change of capacitance on the touch panel. Therefore, the sampling of the TOUCH\_OUT signal is critical. Figure 60.4-2 shows the preprocessing of the TOUCH\_OUT signal.

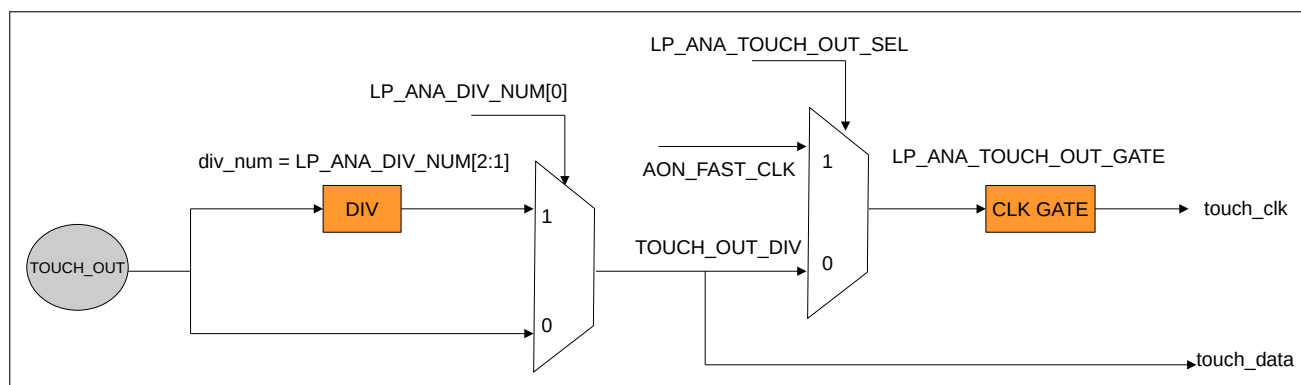


Figure 60.4-2. TOUCH\_OUT Signal Preprocessing

The frequency of TOUCH\_OUT might be much higher than the sampling frequency. Therefore, in order to improve the accuracy of the signal sampling, users can divide the TOUCH\_OUT signal by configuring `LP_ANA_DIV_NUM $n$` .  $n$  in this field represents different frequency modes in frequency hopping (refer to Section 60.4.2.4) with the value 0-2. The lowest bit configures whether to enable or disable frequency division. Set the lowest bit to 0 to disable frequency division, and set it to 1 to enable division. The highest two bits indicate the frequency division coefficient with a value of 0/1/2/3, corresponding to the 0/2/4/6 divisions.

If the frequency is still higher than twice the sampling frequency after frequency division, set `LP_ANA_TOUCH_OUT_SEL` to 1 to directly use the divided TOUCH\_OUT signal as a clock. Set `LP_ANA_TOUCH_OUT_GATE` to configure whether to enable this clock.

### 60.4.2.1 Measurement Process

Depending on whether frequency hopping is enabled, a measurement of a touch pin can be divided into the following two processes:

- Frequency hopping not enabled:

1. The Touch FSM selects the touch sensor to be measured. The relevant signals are routed to that touch sensor.
  2. The Touch FSM drives the START signal to the touch sensor to initiate the measurement. Internally, the Touch FSM starts a touch counter to time the duration of the measurement.
  3. A pulse counter in the Touch FSM is incremented on each pulse received from the touch sensor's TOUCH\_OUT signal.
  4. When the pulse counter reaches the count threshold set in [LP\\_ANA\\_TOUCH\\_MEAS\\_NUM0](#), the measurement is complete. The START signal is de-asserted, and the touch counter is stopped. The value of the stopped touch counter x LP\_DYN\_SLOW\_CLK cycles indicates the time taken to charge and discharge the touch pin for [LP\\_ANA\\_TOUCH\\_MEAS\\_NUM0](#) cycles.
- Frequency hopping enabled:
    1. The Touch FSM selects the touch sensor to be measured. The relevant signals are routed to that touch sensor.
    2. Set [LP\\_ANA\\_FREQ\\_SCAN\\_EN](#) to 1 to enable frequency hopping, configure [LP\\_ANA\\_FREQ\\_SCAN\\_CNT\\_LIMIT](#) to set the number of frequency modes supported for frequency hopping (up to three), and configure the parameters required for each frequency mode and the number of pulses for sampling the TOUCH\_OUT signal.
    3. The Touch FSM drives the START signal to the touch sensor to initiate the measurement. Internally, the Touch FSM starts a touch counter to time the duration of the measurement.
    4. A pulse counter in the Touch FSM is incremented on each pulse received from the touch sensor's TOUCH\_OUT signal.
    5. When the pulse counter reaches the count threshold set in [LP\\_ANA\\_TOUCH\\_MEAS\\_NUM0](#) of the corresponding frequency mode, the measurement is complete. The value of the stopped touch counter indicates the time taken to charge and discharge the touch pin [LP\\_ANA\\_TOUCH\\_MEAS\\_NUM0](#) cycles with the current frequency.
    6. The sampling frequency counter is incremented by 1 and the touch counter is cleared. The next frequency mode will be sampled after a small delay. When the samplings are finished for all frequency modes, the START signal is de-asserted, and the touch counter is cleared.

The sampled value (measured value) is the value of the touch counter indicated as touch\_raw\_data. The value of touch\_raw\_data can be read from [RTC\\_TOUCH\\_PADn\\_DATA](#). Note that the value returned by [RTC\\_TOUCH\\_PADn\\_DATA](#) may also be the filtered touch\_raw\_data, i.e., touch\_smooth\_data and benchmark. Configure [LP\\_ANA\\_TOUCH\\_DATA\\_SEL](#) and [LP\\_ANA\\_TOUCH\\_FREQ\\_SEL](#) to select the type of the return value. For more information on the sampled value types and how to judge touch actions based on the data, refer to Section [60.4.3.1](#).

**Note:**

Note that if the pulse does not reach the set threshold for a long time while the touch counter has reached the timeout threshold set in [LP\\_ANA\\_TOUCH\\_TIMEOUT\\_NUM](#) with [LP\\_ANA\\_TOUCH\\_TIMEOUT\\_EN](#) set as 1, the [RTC\\_TOUCH\\_TIMEOUT\\_INT\\_RAW](#) timeout interrupt will be triggered, indicating a circuit abnormality.

### 60.4.2.2 Trigger Source of Measurement

The Touch FSM initiates a measurement by sending a START signal. The START signal can either be triggered by software, or by a dedicated hardware timer known as the “touch timer”. The use of a touch timer allows periodic measurements to be taken without software intervention.

The touch timer is clocked by LP\_DYN\_SLOW\_CLK and should be configured with a period in number of LP\_DYN\_SLOW\_CLK cycles. The START signal is generated when the touch timer expires. The touch timer is reset when the measurement completes and restarts counting until the next expiration.

- To trigger the START signal by software:
  - Set [LP\\_ANA\\_TOUCH\\_START\\_FORCE](#) to 1 to trigger the START signal by software.
  - Software sets [LP\\_ANA\\_TOUCH\\_START\\_EN](#) to 1 and generates the START signal to initiate a measurement.
- To trigger the START signal by the touch timer:
  - Clear [LP\\_ANA\\_TOUCH\\_START\\_FORCE](#).
  - Configure the wait time before the timer sends the START signal (unit: LP\_DYN\_SLOW\_CLK cycle) via [PMU\\_TOUCH\\_WAIT\\_CYCLES](#).
  - Configure the wait time after the timer receives the DONE signal upon the completion of the touch sensor sampling (unit: LP\_DYN\_SLOW\_CLK cycle) via [PMU\\_TOUCH\\_SLEEP\\_CYCLES](#).
  - Set [PMU\\_TOUCH\\_SLEEP\\_TIMER\\_EN](#) to 1 to enable the touch timer.
- To force exit the current measurement cycle of the touch timer by software, there are two approaches:
  1. Generate the end signal of the touch sensor to exit the current measurement cycle of the touch timer.
    - Set [LP\\_ANA\\_TOUCH\\_DONE\\_FORCE](#) to 1 to select the DONE signal to be triggered by software.
    - Software sets [LP\\_ANA\\_TOUCH\\_DONE\\_EN](#) to 1 to generate the DONE signal to end a measurement.
  2. Use the software to directly force the touch timer to exit the current measurement cycle.
    - Set [PMU\\_TOUCH\\_FORCE\\_DONE](#) to generate the DONE signal and force the touch timer to exit the current measurement cycle.

**Note:**

The touch timer waits for a period of time before generating the START signal and after receiving the DONE signal, so the interval between two consecutive measurements is ([PMU\\_TOUCH\\_WAIT\\_CYCLES](#) + [PMU\\_TOUCH\\_SLEEP\\_CYCLES](#)) and the timing clock is LP\_DYN\_SLOW\_CLK.

### 60.4.2.3 Scan Mode

Scan mode involves the Touch FSM taking measurements of multiple touch sensors in sequential order. On every START signal, a new touch sensor is selected for measurement, thus allowing multiple touch pins to be monitored. The scan process is illustrated in Figure [60.4-3](#).

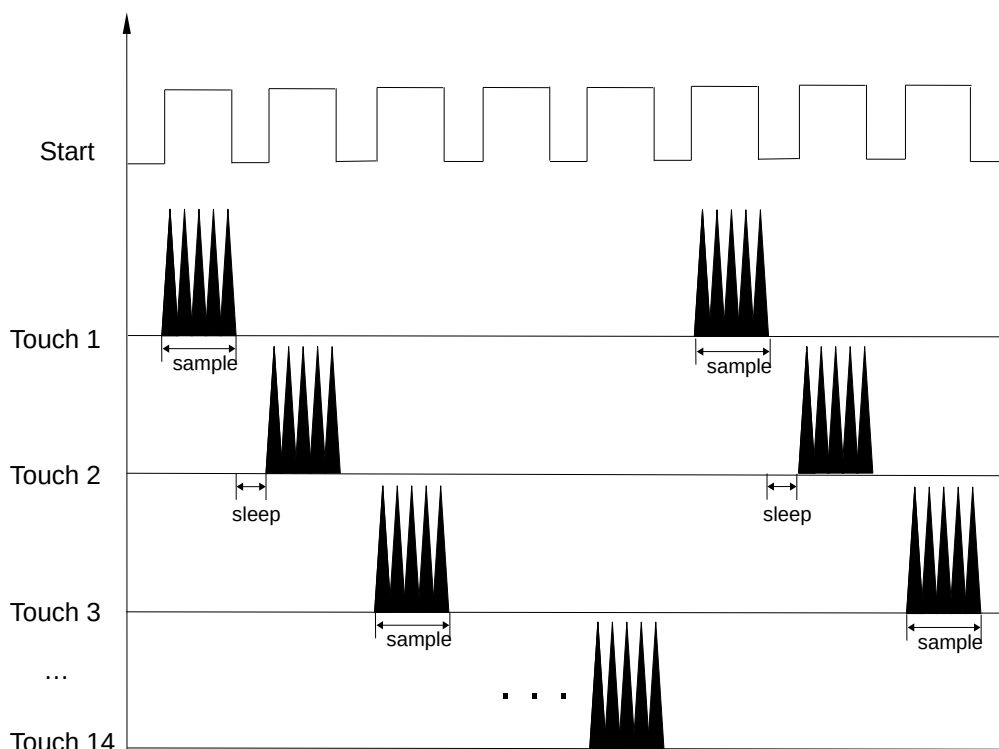


Figure 60.4-3. Timing Diagram of Scan Mode

To enable the scan mode, specify the bit map of the enabled touch pins in [LP\\_ANA\\_TOUCH\\_SCAN\\_PAD\\_MAP](#). The Touch FSM will select one touch pin from the bit map of the enabled touch pins in turn to measure per START signal, i.e., only one measurement is conducted per START signal. As multiple START signals are generated, the Touch FSM will cycle through the enabled touch pins in sequential order.

When measuring for different touch sensors, the sleep interval between two measurements is  $PMU\_TOUCH\_WAIT\_CYCLES + PMU\_TOUCH\_SLEEP\_CYCLES$  with the clock cycle of  $LP\_DYN\_SLOW\_CLK$ .

**Note:**

- After the touch sensor receives the START signal from the touch timer, it waits a short period before starting the measurement to ensure the stability of the sampling signal. The wait time is configured by [LP\\_ANA\\_TOUCH\\_XPD\\_WAIT](#). The timing clock is  $AON\_FAST\_CLK$ .
- If the touch timer is used to generate the START signal, the scanning can be conducted without software intervention.

#### 60.4.2.4 Frequency Hopping

When frequency hopping is enabled, the Touch FSM can measure the touch sensor at different frequency modes. Up to three frequency modes are supported, each corresponding to a set of configuration parameters. When a START signal is generated, the measurement will be carried out for each frequency mode sequentially, and the DONE signal will be generated when the measurements are completed for all frequency modes. Frequency hopping can be enabled together with the scan mode. In such case, users can monitor the touch condition of multiple touch sensors under different frequency modes by generating multiple START

signals. The scanning process is shown in Figure 60.4-4.

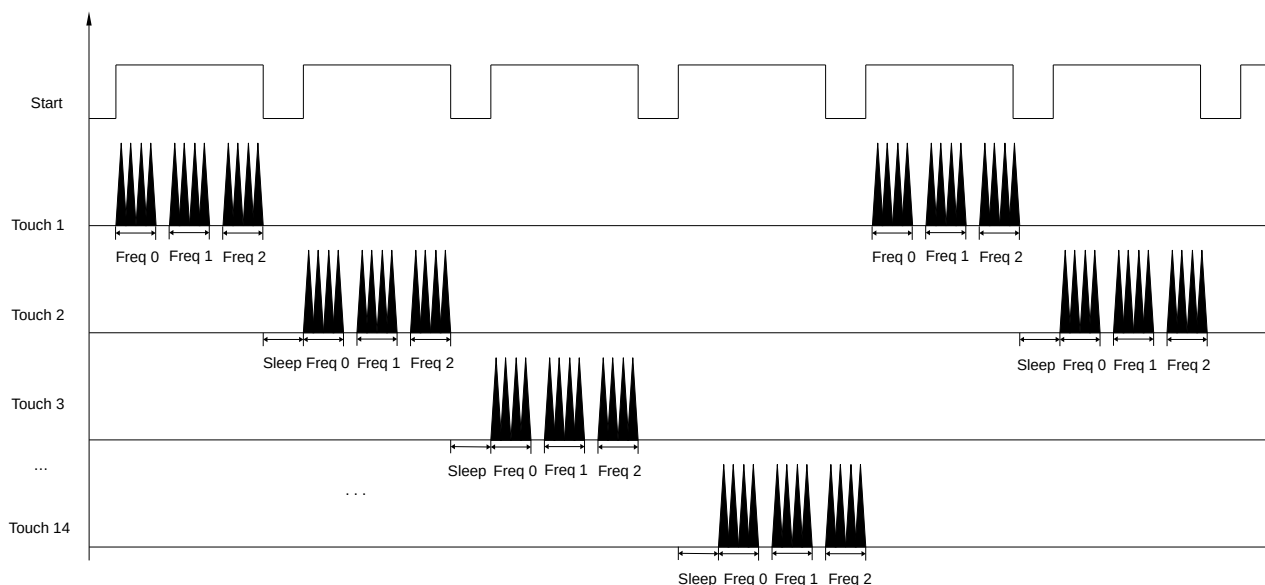


Figure 60.4-4. Touch Sensor Frequency Hopping Sequence Timing

Set `LP_ANA_FREQ_SCAN_EN` to 1 to enable frequency hopping, configure `LP_ANA_FREQ_SCAN_CNT_LIMIT` to select the maximum number of frequency mode supported, and configure the following analog parameter registers to adjust the sampling performance of the touch sensor at different frequency modes:

- `LP_ANA_TOUCH_FREQ $n$ _DBIAS` ( $n = 0-2$ )
- `LP_ANA_TOUCH_FREQ $n$ _DRV_HS` ( $n = 0-2$ )
- `LP_ANA_TOUCH_FREQ $n$ _DRV_LS` ( $n = 0-2$ )
- `LP_ANA_TOUCH_FREQ $n$ _DRES_LPF` ( $n = 0-2$ )
- `LP_ANA_TOUCH_FREQ $n$ _DCAP_LPF` ( $n = 0-2$ )

Upon each START signal, the Touch FSM measures consecutively for frequency mode 0, 1, and 2. A START signal triggers a maximum of three measurement operations. When the measurements are completed for all frequency modes, the START signal is de-asserted until the next measurement. Therefore, the Touch FSM performs measurements on the enabled touch sensors in sequence upon multiple START signals. A touch on the same sensor is recognized only when the number of hit frequency points is greater than or equal to `LP_ANA_FREQ_SCAN_CNT_RISE`.

**Note:**

- When the same touch sensor is measured at different frequency modes, the wait time between measurements can be configured by `LP_ANA_TOUCH_XPD_WAIT` (count clock: `AON_FAST_CLK`).
- After receiving the START signal from the touch timer, the touch sensor waits for a short period of time before starting measuring to ensure the stability of the sampling signal. The wait time can be configured by `LP_ANA_TOUCH_XPD_WAIT` (count clock: `AON_FAST_CLK`).
- If a touch timer is used to generate the START signal, the scanning can be conducted without software intervention.

## 60.4.3 Touch Detection

### 60.4.3.1 Sampled Value

As introduced in Section 60.4.2.1, the sampled value of a measurement can be read from [RTC\\_TOUCH\\_PAD \$n\$ \\_DATA](#). However, the specific type of sampled value stored in [RTC\\_TOUCH\\_PAD \$n\$ \\_DATA](#) is determined together by [LP\\_ANA\\_TOUCH\\_DATA\\_SEL](#) and [LP\\_ANA\\_TOUCH\\_FREQ\\_SEL](#). The following three sampled value types are supported:

1. **touch\_raw\_data**: When the measurement is completed, the touch counter's value indicates the time taken to charge and discharge the touch pin a fixed number of times in LP\_DYN\_SLOW\_CLK cycles. After the IIR filter is turned on, the data is the filtered data.
2. **touch\_smooth\_data**: A series of moving averages generated by the touch\_raw\_data values sampled from a single touch sensor filtered by an IIR filter.

touch\_smooth\_data is less prone to noise spikes or outlier samples, thus often used for hardware touch detection. If frequency hopping is enabled, each touch pin will have a separate set of smooth\_touch\_data at different frequency modes and do not affect each other. The IIR filter that generates touch\_smooth\_data is configured via [LP\\_ANA\\_TOUCH\\_SMOOTH\\_LVL](#). See Table 60.4-1 for more information.

Table 60.4-1. touch\_smooth\_data Calculation

| <a href="#">LP_ANA_TOUCH_SMOOTH_LVL</a> | Type    | Formula                                                          |
|-----------------------------------------|---------|------------------------------------------------------------------|
| 0                                       | —       | touch_raw_data                                                   |
| 1                                       | IIR 1/2 | $1/2 \text{ touch\_raw\_data} + 1/2 \text{ touch\_smooth\_data}$ |
| 2                                       | IIR 1/4 | $1/4 \text{ touch\_raw\_data} + 3/4 \text{ touch\_smooth\_data}$ |
| 3                                       | IIR 1/8 | $1/8 \text{ touch\_raw\_data} + 7/8 \text{ touch\_smooth\_data}$ |

3. **benchmark**: Also generated from the touch\_raw\_data values by IIR filtering. However, the moving average window for benchmark is much wider.

The benchmark value is usually used to represent the stable reading of a touch pin without the effect from a touch action. The touch detection is based on the benchmark value. Each touch sensor has a separate set of benchmark values at different frequency modes with a default value of 0xFFFF. The IIR filter that generates benchmark is configured via [LP\\_ANA\\_TOUCH\\_FILTER\\_MODE](#). Refer to Table 60.4-2 for more information.

In addition to the IIR filtering, users can also update the benchmark value directly by software. The software update procedure is as follows:

- Configure the target frequency point to be updated through [LP\\_ANA\\_TOUCH\\_UPDATE\\_BENCHMARK\\_FREQ\\_SEL](#).
- Configure the target channel to be updated through [LP\\_ANA\\_TOUCH\\_UPDATE\\_BENCHMARK\\_PAD\\_SEL](#).
- Configure the benchmark data value through [LP\\_ANA\\_TOUCH\\_BENCHMARK\\_SW](#).
- Configure [LP\\_ANA\\_TOUCH\\_UPDATE\\_BENCHMARK\\_SW](#) to trigger the benchmark data update by software.

This method is generally used for software debugging. When the benchmark data of the target frequency point and channel has been updated, the interrupt [TOUCH\\_BENCHMARK\\_UPDATE\\_INT](#) will be generated.

Table 60.4-2. benchmark Algorithm

| <a href="#">LP_ANA_TOUCH_FILTER_MODE</a> | Type      | Formula                                                           |
|------------------------------------------|-----------|-------------------------------------------------------------------|
| 0                                        | IIR 1/4   | $1/4 \text{ touch\_raw\_data} + 3/4 \text{ benchmark}$            |
| 1                                        | IIR 1/8   | $1/8 \text{ touch\_raw\_data} + 7/8 \text{ benchmark}$            |
| 2                                        | IIR 1/16  | $1/16 \text{ touch\_raw\_data} + 15/16 \text{ benchmark}$         |
| 3                                        | IIR 1/32  | $1/32 \text{ touch\_raw\_data} + 31/32 \text{ benchmark}$         |
| 4                                        | IIR 1/64  | $1/64 \text{ touch\_raw\_data} + 63/64 \text{ benchmark}$         |
| 5                                        | IIR 1/128 | $1/128 \text{ touch\_raw\_data} + 127/128 \text{ benchmark}$      |
| 6                                        | IIR 1/256 | $1/256 \text{ touch\_raw\_data} + 255/256 \text{ benchmark}$      |
| 7                                        | JITTER    | $\text{touch\_raw\_data} \pm \text{LP\_ANA\_TOUCH\_JITTER\_STEP}$ |

### 60.4.3.2 Hardware Touch Detection

In hardware touch detection, the touch sensor can detect finger touch or release and trigger an interrupt. To use hardware touch detection, the following key parameters need to be defined:

- **finger\_threshold:** The threshold value used to determine a touch or a touch interruption. [finger\\_threshold](#) is configured via [LP\\_ANA\\_TOUCH\\_PADx\\_THy](#), where *x* (1-14) represents the touch pin index and *y* (0-2) represents the frequency mode configured in hopping frequency.
- **noise\_threshold:** Indicates the maximum threshold of the upward and downward fluctuation of [touch\\_smooth\\_data](#). [noise\\_threshold](#) is configured via [LP\\_ANA\\_TOUCH\\_NOISE\\_THRES](#).
- **hysteresis:** Used to avoid touch misjudgments caused by fluctuations of [touch\\_smooth\\_data](#) around touch thresholds. The hysteresis value is configured via [LP\\_ANA\\_TOUCH\\_HHYSTERESIS](#).

The [finger\\_threshold](#) and [noise\\_threshold](#) values are not absolute thresholds, but offsets from the benchmark value, while the hysteresis value is an offset from [finger\\_threshold](#). This prevents from misdiagnosing the gradual or slow changes of [touch\\_raw\\_data](#) caused by environmental factors such as temperature, power, or noise as a touch action.

After a series of calculations and comparisons based on the above parameters, the touch sensor will output the touch results detected on the touch pins via interrupts. Each touch sensor corresponds to a control register [LP\\_ANA\\_TOUCH\\_OUTEN](#) to configure whether to output the detection result.

### 60.4.4 Proximity Mode

When an object such as a finger is close to, but not touching, a touch pin, a small change in capacitance occurs on the touch pin, which is much smaller than the change caused by a physical touch. Proximity mode allows for the detection of these small changes.

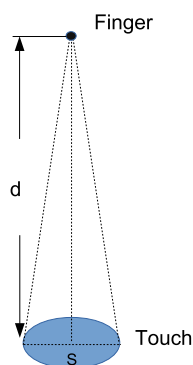


Figure 60.4-5. Sensing Area

- The maximum detection distance “d” is 16 cm and the sensing area “S” is 20 cm<sup>2</sup>, where “d” is positively correlated with “S” as shown in Figure 60.4-5.
- Up to three touch pins can be configured to operate simultaneously in proximity mode.
- Frequency hopping is also supported in proximity mode.

In proximity mode, a touch sensor samples for a fixed number of times and accumulates the sample values. If the final cumulative value exceeds the configured threshold, a proximity object is considered to be detected and an interrupt will be triggered. Note that since the sample values are accumulated, the values introduced in Section 60.4.3.1, i.e., touch\_raw\_data, touch\_smooth\_data, and benchmark, will not be generated in proximity mode.

To operate in proximity mode:

- Configure a touch sensor to operate in proximity mode by setting `LP_ANA_TOUCH_APPROACH_PAD0`, `LP_ANA_TOUCH_APPROACH_PAD1`, or `LP_ANA_TOUCH_APPROACH_PAD2`.
- Configure `LP_ANA_TOUCH_APPROACH_LIMIT` to adjust the number of samplings to generate the cumulative value. The touch sensor contains an internal sample counter to record the number of samplings.
- Set the threshold value via `LP_ANA_TOUCH_PADx_THy`, where *x* (1-14) represents the touch pin index and *y* (0-2) represents the frequency mode configured in hopping frequency.
- When the sample counter reaches the number of samplings configured in `LP_ANA_TOUCH_APPROACH_MEAS_NUMn`:
  - If the cumulative value exceeds the threshold value, an interrupt will be triggered.
  - The sample counter and the cumulative value are reset to 0. The touch sensor restarts accumulating the sample values.

### 60.4.5 Sleep Mode

To reduce power consumption, the touch sensor also supports sleep mode. In sleep mode, users can designate one touch sensor to operate actively (hereinafter referred to as the sleeping touch sensor) to monitor whether a touch occurs in real time, while the remaining 13 touch sensors are all powered down in sleep mode. Once the sleeping touch sensor detects a touch or a touch release, it will generate an interrupt to wake up the rest of the touch sensors. The sleeping touch sensor also supports frequency hopping and

proximity mode. There is a separate set of threshold parameters for the sleeping touch sensor, which can be configured as follows:

- Configure [LP\\_ANA\\_TOUCH\\_SLP\\_PAD](#) to select a particular touch sensor as the sleeping touch sensor.
- Configure [LP\\_ANA\\_TOUCH\\_SLP\\_TH0](#), [LP\\_ANA\\_TOUCH\\_SLP\\_TH1](#), and [LP\\_ANA\\_TOUCH\\_SLP\\_TH2](#) to set the touch threshold for the sleeping touch sensor at different frequency modes.
- Configure [LP\\_ANA\\_TOUCH\\_SLP\\_APPROACH\\_EN](#) to enable the proximity mode for the sleeping touch sensor.
- Configure [LP\\_ANA\\_TOUCH\\_SLP\\_CHANNEL\\_CLR](#) to clear the intermediate data for the sleeping touch sensor, such as benchmark, touch\_raw\_data, and the number of sampling.

**Note:**

The configuration for the noise\_threshold, hysteresis, touch\_smooth\_data, and benchmark values are the same for the sleeping touch sensor in sleep mode as in the normal mode.

## 60.4.6 Moisture Tolerance

If there are water droplets on the touch pads and the droplets are large enough to physically bridge two more adjacent touch pads, the adjacent touch pads may be electrically coupled. Coupled touch pads can cause false touch detection due to capacitance changes caused by the coupling. If a water droplet is connected across the touch pad and ground, a large parasitic capacitance will be introduced, also leading to false detection. The moisture tolerance feature can mitigate the impact of water droplets.

To configure moisture tolerance:

- Configure [LP\\_ANA\\_TOUCH\\_SHIELD\\_PAD\\_EN](#) to enable moisture tolerance.
- Configure [LP\\_ANA\\_TOUCH\\_BUFSEL](#) to select a touch pin to be used for moisture tolerance.

**Note:**

The moisture tolerance feature only works at frequencies below 10 MHz. Therefore, when the frequency is higher than 10 MHz, please disable the moisture tolerance feature.

## 60.4.7 Water Rejection

If the sensor array becomes wet, i.e., the majority of the sensor pads are soaked by water, most (if not all) of the touch pads will become unusable due to the possible false touch detections. ESP32-P4 supports the water rejection feature to shut down the sensor array if it is detected to be wet.

Configure [LP\\_ANA\\_TOUCH\\_OUT\\_RING](#) to select one of the touch pins to be used for the water rejection feature.

## 60.5 Interrupts

ESP32-P4's touch sensor can generate the LP\_TOUCH\_INTR signal that will be sent to the [Interrupt Matrix](#).

There are several internal interrupt sources from touch sensor that can generate the LP\_TOUCH\_INTR signal. The interrupt sources from touch sensor are listed with their trigger conditions and the resulted interrupt signal in Table 60.5-1.

Table 60.5-1. Touch Sensor's Internal Interrupt Sources

| Internal Interrupt Source    | Trigger Condition                                                                         | Interrupt Signal |
|------------------------------|-------------------------------------------------------------------------------------------|------------------|
| TOUCH_APPROACH_LOOP_DONE_INT | Completion of proximity mode cumulative sampling                                          | LP_TOUCH_INTR    |
| TOUCH_TIMEOUT_INT            | Timeout for the TOUCH_OUT signal sampling                                                 |                  |
| TOUCH_INACTIVE_INT           | Touch release detected                                                                    |                  |
| TOUCH_ACTIVE_INT             | Touch detected                                                                            |                  |
| TOUCH_DONE_INT               | Completion of sampling an individual touch pin at a certain frequency mode for comparison |                  |
| TOUCH_SCAN_DONE_INT          | Completion of sampling selected touch pins for comparison at all frequency modes          |                  |
| TOUCH_BENCHMARK_UPDATE_INT   | Completion of software benchmark data update                                              |                  |

**Note:**

For definitions of *interrupt*, *interrupt signal*, *interrupt source*, and their correlations, please refer to Chapter [12 Interrupt Matrix](#) > Section [12.2 Interrupt Terminology in ESP32-P4](#).

Each interrupt source can be configured by a common set of registers that are described in Section [Interrupt Configuration Registers](#). The specific registers can be found in Section [60.6 Register Summary](#).

# 60.6 Register Summary

## 60.6.1 Interrupt and Status Register Summary

The addresses in this section are relative to Touch Sensor base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

| Name                                     | Description                            | Address | Access   |
|------------------------------------------|----------------------------------------|---------|----------|
| <b>Interrupt Register</b>                |                                        |         |          |
| <a href="#">RTC_TOUCH_INT_RAW_REG</a>    | Touch sensor interrupt raw register    | 0x0000  | R/SS/WTC |
| <a href="#">RTC_TOUCH_INT_ST_REG</a>     | Touch sensor interrupt status register | 0x0004  | RO       |
| <a href="#">RTC_TOUCH_INT_ENA_REG</a>    | Touch sensor interrupt enable register | 0x0008  | R/W      |
| <a href="#">RTC_TOUCH_INT_CLR_REG</a>    | Touch sensor interrupt clear register  | 0x000C  | WT       |
| <b>Status Register</b>                   |                                        |         |          |
| <a href="#">RTC_TOUCH_CHN_STATUS_REG</a> | Scan mode status register              | 0x0010  | RO       |
| <a href="#">RTC_TOUCH_STATUS_1_REG</a>   | Touch pin 1 status register            | 0x0018  | RO       |
| <a href="#">RTC_TOUCH_STATUS_2_REG</a>   | Touch pin 2 status register            | 0x001C  | RO       |
| <a href="#">RTC_TOUCH_STATUS_3_REG</a>   | Touch pin 3 status register            | 0x0020  | RO       |
| <a href="#">RTC_TOUCH_STATUS_4_REG</a>   | Touch pin 4 status register            | 0x0024  | RO       |
| <a href="#">RTC_TOUCH_STATUS_5_REG</a>   | Touch pin 5 status register            | 0x0028  | RO       |
| <a href="#">RTC_TOUCH_STATUS_6_REG</a>   | Touch pin 6 status register            | 0x002C  | RO       |
| <a href="#">RTC_TOUCH_STATUS_7_REG</a>   | Touch pin 7 status register            | 0x0030  | RO       |
| <a href="#">RTC_TOUCH_STATUS_8_REG</a>   | Touch pin 8 status register            | 0x0034  | RO       |
| <a href="#">RTC_TOUCH_STATUS_9_REG</a>   | Touch pin 9 status register            | 0x0038  | RO       |
| <a href="#">RTC_TOUCH_STATUS_10_REG</a>  | Touch pin 10 status register           | 0x003C  | RO       |
| <a href="#">RTC_TOUCH_STATUS_11_REG</a>  | Touch pin 11 status register           | 0x0040  | RO       |
| <a href="#">RTC_TOUCH_STATUS_12_REG</a>  | Touch pin 12 status register           | 0x0044  | RO       |
| <a href="#">RTC_TOUCH_STATUS_13_REG</a>  | Touch pin 13 status register           | 0x0048  | RO       |
| <a href="#">RTC_TOUCH_STATUS_14_REG</a>  | Touch pin 14 status register           | 0x004C  | RO       |
| <a href="#">RTC_TOUCH_STATUS_15_REG</a>  | Sleep mode touch pin status register   | 0x0050  | RO       |

| Name                                         | Description                       | Address | Access |
|----------------------------------------------|-----------------------------------|---------|--------|
| <a href="#">RTC_TOUCH_STATUS_16_REG</a>      | Proximity mode status register    | 0x0054  | RO     |
| <a href="#">RTC_TOUCH_STATUS_17_REG</a>      | Frequency hopping status register | 0x0058  | RO     |
| <a href="#">RTC_TOUCH_CHN_TMP_STATUS_REG</a> | Touch status register             | 0x005C  | RO     |
| <b>Version Control Register</b>              |                                   |         |        |
| <a href="#">RTC_TOUCH_DATE_REG</a>           | Version control register          | 0x0100  | R/W    |

## 60.6.2 Configuration Register Summary

The addresses in this section are relative to LP Analog Peripheral base address provided in Table [7.3-2](#) in Chapter [7 System and Memory](#).

The abbreviations given in Column **Access** are explained in Section [Access Types for Registers](#).

| Name                                                    | Description                                                                    | Address | Access |
|---------------------------------------------------------|--------------------------------------------------------------------------------|---------|--------|
| <b>Configuration Register</b>                           |                                                                                |         |        |
| <a href="#">LP_ANA_TOUCH_APPROACH_WORK_MEAS_NUM_REG</a> | Configuration register for the number of measurements in proximity mode        | 0x00FC  | R/W    |
| <a href="#">LP_ANA_TOUCH_SCAN_CTRL1_REG</a>             | Scan mode configuration register 1                                             | 0x0100  | R/W    |
| <a href="#">LP_ANA_TOUCH_SCAN_CTRL2_REG</a>             | Scan mode configuration register 2                                             | 0x0104  | R/W    |
| <a href="#">LP_ANA_TOUCH_WORK_REG</a>                   | Sample preprocessing register                                                  | 0x0108  | varies |
| <a href="#">LP_ANA_TOUCH_WORK_MEAS_NUM_REG</a>          | Configuration register for number of measurements at different frequency modes | 0x010C  | R/W    |
| <a href="#">LP_ANA_TOUCH_FILTER1_REG</a>                | Touch detection configuration register 1                                       | 0x0110  | R/W    |
| <a href="#">LP_ANA_TOUCH_FILTER2_REG</a>                | Touch detection configuration register 2                                       | 0x0114  | R/W    |
| <a href="#">LP_ANA_TOUCH_FILTER3_REG</a>                | Touch detection configuration register 3                                       | 0x0118  | varies |
| <a href="#">LP_ANA_TOUCH_SLPO_REG</a>                   | Sleep mode configuration register 1                                            | 0x011C  | varies |
| <a href="#">LP_ANA_TOUCH_SLP1_REG</a>                   | Sleep mode configuration register 2                                            | 0x0120  | R/W    |
| <a href="#">LP_ANA_TOUCH_CLR_REG</a>                    | Status clear register                                                          | 0x0124  | WT     |
| <a href="#">LP_ANA_TOUCH_APPROACH_REG</a>               | Proximity mode configuration register                                          | 0x0128  | R/W    |
| <a href="#">LP_ANA_TOUCH_FREQ0_SCAN_PARA_REG</a>        | Analog parameter configuration register for touch sensor for frequency mode 0  | 0x012C  | R/W    |
| <a href="#">LP_ANA_TOUCH_FREQ1_SCAN_PARA_REG</a>        | Analog parameter configuration register for touch sensor for frequency mode 1  | 0x0130  | R/W    |

| Name                                             | Description                                                                   | Address | Access |
|--------------------------------------------------|-------------------------------------------------------------------------------|---------|--------|
| <a href="#">LP_ANA_TOUCH_FREQ2_SCAN_PARA_REG</a> | Analog parameter configuration register for touch sensor for frequency mode 2 | 0x0134  | R/W    |
| <a href="#">LP_ANA_TOUCH_ANA_PARA_REG</a>        | Touch sensor analog parameter configuration register                          | 0x0138  | R/W    |
| <a href="#">LP_ANA_TOUCH_MUX0_REG</a>            | DEBUG mode selection configuration register                                   | 0x013C  | R/W    |
| <a href="#">LP_ANA_TOUCH_MUX1_REG</a>            | DEBUG mode function configuration register                                    | 0x0140  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD1_TH0_REG</a>        | Touch threshold configuration register for touch pin 1 for frequency mode 0   | 0x0150  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD1_TH1_REG</a>        | Touch threshold configuration register for touch pin 1 for frequency mode 1   | 0x0154  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD1_TH2_REG</a>        | Touch threshold configuration register for touch pin 1 for frequency mode 2   | 0x0158  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD2_TH0_REG</a>        | Touch threshold configuration register for touch pin 2 for frequency mode 0   | 0x015C  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD2_TH1_REG</a>        | Touch threshold configuration register for touch pin 2 for frequency mode 1   | 0x0160  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD2_TH2_REG</a>        | Touch threshold configuration register for touch pin 2 for frequency mode 2   | 0x0164  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD3_TH0_REG</a>        | Touch threshold configuration register for touch pin 3 for frequency mode 0   | 0x0168  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD3_TH1_REG</a>        | Touch threshold configuration register for touch pin 3 for frequency mode 1   | 0x016C  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD3_TH2_REG</a>        | Touch threshold configuration register for touch pin 3 for frequency mode 2   | 0x0170  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD4_TH0_REG</a>        | Touch threshold configuration register for touch pin 4 for frequency mode 0   | 0x0174  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD4_TH1_REG</a>        | Touch threshold configuration register for touch pin 4 for frequency mode 1   | 0x0178  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD4_TH2_REG</a>        | Touch threshold configuration register for touch pin 4 for frequency mode 2   | 0x017C  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD5_TH0_REG</a>        | Touch threshold configuration register for touch pin 5 for frequency mode 0   | 0x0180  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD5_TH1_REG</a>        | Touch threshold configuration register for touch pin 5 for frequency mode 1   | 0x0184  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD5_TH2_REG</a>        | Touch threshold configuration register for touch pin 5 for frequency mode 2   | 0x0188  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD6_TH0_REG</a>        | Touch threshold configuration register for touch pin 6 for frequency mode 0   | 0x018C  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD6_TH1_REG</a>        | Touch threshold configuration register for touch pin 6 for frequency mode 1   | 0x0190  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD6_TH2_REG</a>        | Touch threshold configuration register for touch pin 6 for frequency mode 2   | 0x0194  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD7_TH0_REG</a>        | Touch threshold configuration register for touch pin 7 for frequency mode 0   | 0x0198  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD7_TH1_REG</a>        | Touch threshold configuration register for touch pin 7 for frequency mode 1   | 0x019C  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD7_TH2_REG</a>        | Touch threshold configuration register for touch pin 7 for frequency mode 2   | 0x01A0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD8_TH0_REG</a>        | Touch threshold configuration register for touch pin 8 for frequency mode 0   | 0x01A4  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD8_TH1_REG</a>        | Touch threshold configuration register for touch pin 8 for frequency mode 1   | 0x01A8  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD8_TH2_REG</a>        | Touch threshold configuration register for touch pin 8 for frequency mode 2   | 0x01AC  | R/W    |

| Name                                       | Description                                                                  | Address | Access |
|--------------------------------------------|------------------------------------------------------------------------------|---------|--------|
| <a href="#">LP_ANA_TOUCH_PAD9_TH0_REG</a>  | Touch threshold configuration register for touch pin 9 for frequency mode 0  | 0x01B0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD9_TH1_REG</a>  | Touch threshold configuration register for touch pin 9 for frequency mode 1  | 0x01B4  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD9_TH2_REG</a>  | Touch threshold configuration register for touch pin 9 for frequency mode 2  | 0x01B8  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD10_TH0_REG</a> | Touch threshold configuration register for touch pin 10 for frequency mode 0 | 0x01BC  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD10_TH1_REG</a> | Touch threshold configuration register for touch pin 10 for frequency mode 1 | 0x01C0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD10_TH2_REG</a> | Touch threshold configuration register for touch pin 10 for frequency mode 2 | 0x01C4  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD11_TH0_REG</a> | Touch threshold configuration register for touch pin 11 for frequency mode 0 | 0x01C8  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD11_TH1_REG</a> | Touch threshold configuration register for touch pin 11 for frequency mode 1 | 0x01CC  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD11_TH2_REG</a> | Touch threshold configuration register for touch pin 11 for frequency mode 2 | 0x01D0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD12_TH0_REG</a> | Touch threshold configuration register for touch pin 12 for frequency mode 0 | 0x01D4  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD12_TH1_REG</a> | Touch threshold configuration register for touch pin 12 for frequency mode 1 | 0x01D8  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD12_TH2_REG</a> | Touch threshold configuration register for touch pin 12 for frequency mode 2 | 0x01DC  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD13_TH0_REG</a> | Touch threshold configuration register for touch pin 13 for frequency mode 0 | 0x01E0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD13_TH1_REG</a> | Touch threshold configuration register for touch pin 13 for frequency mode 1 | 0x01E4  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD13_TH2_REG</a> | Touch threshold configuration register for touch pin 13 for frequency mode 2 | 0x01E8  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD14_TH0_REG</a> | Touch threshold configuration register for touch pin 14 for frequency mode 0 | 0x01EC  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD14_TH1_REG</a> | Touch threshold configuration register for touch pin 14 for frequency mode 1 | 0x01F0  | R/W    |
| <a href="#">LP_ANA_TOUCH_PAD14_TH2_REG</a> | Touch threshold configuration register for touch pin 14 for frequency mode 2 | 0x01F4  | R/W    |
| <a href="#">LP_ANA_DATE_REG</a>            | Version control register                                                     | 0x03FC  | R/W    |

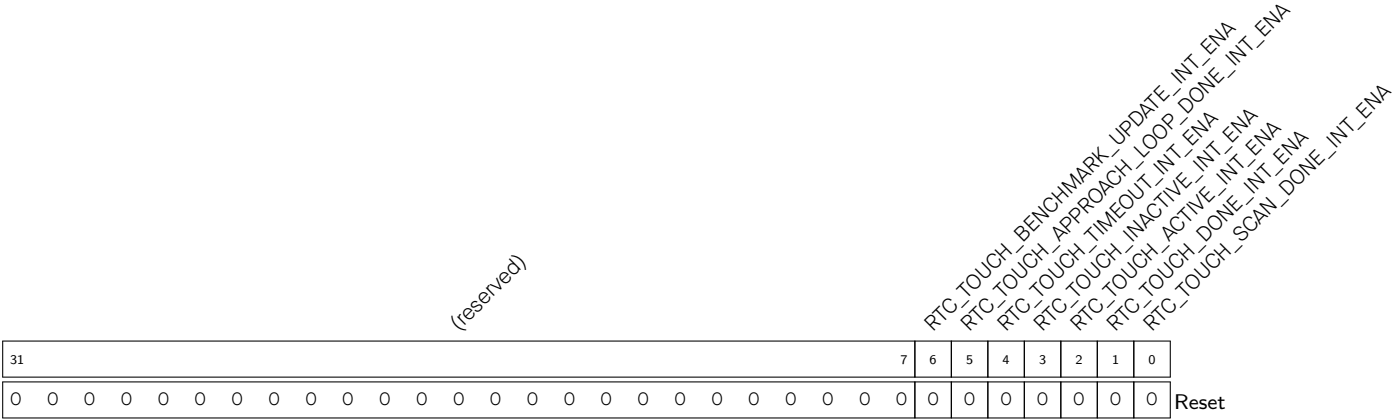


Register 60.2. RTC\_TOUCH\_INT\_ST\_REG (0x0004)

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (reserved) |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_BENCHMARK_UPDATE_INT_ST   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_APPROACH_LOOP_DONE_INT_ST |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_TIMEOUT_INT_ST            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_INACTIVE_INT_ST           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_ACTIVE_INT_ST             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_DONE_INT_ST               |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | RTC_TOUCH_SCAN_DONE_INT_ST          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31         |   |   |   |   |   |   | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |   |   |   |   |   |   |   |   |   |                                     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0                                   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

- RTC\_TOUCH\_SCAN\_DONE\_INT\_ST The masked interrupt status of TOUCH\_DONE\_INT. (RO)
- RTC\_TOUCH\_DONE\_INT\_ST The masked interrupt status of TOUCH\_DONE\_INT. (RO)
- RTC\_TOUCH\_ACTIVE\_INT\_ST The masked interrupt status of TOUCH\_ACTIVE\_INT. (RO)
- RTC\_TOUCH\_INACTIVE\_INT\_ST The masked interrupt status of TOUCH\_INACTIVE\_INT. (RO)
- RTC\_TOUCH\_TIMEOUT\_INT\_ST The masked interrupt status of TOUCH\_TIMEOUT\_INT. (RO)
- RTC\_TOUCH\_APPROACH\_LOOP\_DONE\_INT\_ST The masked interrupt status of TOUCH\_APPROACH\_LOOP\_DONE\_INT. (RO)
- RTC\_TOUCH\_BENCHMARK\_UPDATE\_INT\_ST The masked interrupt status of TOUCH\_BENCHMARK\_UPDATE\_INT. (RO)

Register 60.3. RTC\_TOUCH\_INT\_ENA\_REG (0x0008)



- RTC\_TOUCH\_SCAN\_DONE\_INT\_ENA Write 1 to enable [TOUCH\\_DONE\\_INT](#). (R/W)
- RTC\_TOUCH\_DONE\_INT\_ENA Write 1 to enable [TOUCH\\_DONE\\_INT](#). (R/W)
- RTC\_TOUCH\_ACTIVE\_INT\_ENA Write 1 to enable [TOUCH\\_ACTIVE\\_INT](#). (R/W)
- RTC\_TOUCH\_INACTIVE\_INT\_ENA Write 1 to enable [TOUCH\\_INACTIVE\\_INT](#). (R/W)
- RTC\_TOUCH\_TIMEOUT\_INT\_ENA Write 1 to enable [TOUCH\\_TIMEOUT\\_INT](#). (R/W)
- RTC\_TOUCH\_APPROACH\_LOOP\_DONE\_INT\_ENA Write 1 to enable [TOUCH\\_APPROACH\\_LOOP\\_DONE\\_INT](#). (R/W)
- RTC\_TOUCH\_BENCHMARK\_UPDATE\_INT\_ENA Write 1 to enable [TOUCH\\_BENCHMARK\\_UPDATE\\_INT](#). (R/W)

## Register 60.4. RTC\_TOUCH\_INT\_CLR\_REG (0x000C)

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (reserved) |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

RTC\_TOUCH\_SCAN\_DONE\_INT\_CLR Write 1 to clear [TOUCH\\_DONE\\_INT](#). (WT)

RTC\_TOUCH\_DONE\_INT\_CLR Write 1 to clear [TOUCH\\_DONE\\_INT](#). (WT)

RTC\_TOUCH\_ACTIVE\_INT\_CLR Write 1 to clear [TOUCH\\_ACTIVE\\_INT](#). (WT)

RTC\_TOUCH\_INACTIVE\_INT\_CLR Write 1 to clear [TOUCH\\_INACTIVE\\_INT](#). (WT)

RTC\_TOUCH\_TIMEOUT\_INT\_CLR Write 1 to clear [TOUCH\\_TIMEOUT\\_INT](#). (WT)

RTC\_TOUCH\_APPROACH\_LOOP\_DONE\_INT\_CLR Write 1 to clear [TOUCH\\_APPROACH\\_LOOP\\_DONE\\_INT](#).  
(WT)

RTC\_TOUCH\_BENCHMARK\_UPDATE\_INT\_CLR Write 1 to clear [TOUCH\\_BENCHMARK\\_UPDATE\\_INT](#).  
(WT)

Register 60.5. RTC\_TOUCH\_CHN\_STATUS\_REG (0x0010)

|            |   |   |   |   |   |   |   |   |   |   |    |                     |    |      |    |                     |  |  |  |                      |  |  |  |       |  |   |
|------------|---|---|---|---|---|---|---|---|---|---|----|---------------------|----|------|----|---------------------|--|--|--|----------------------|--|--|--|-------|--|---|
| (reserved) |   |   |   |   |   |   |   |   |   |   |    | RTC_TOUCH_SCAN_CURR |    |      |    | RTC_TOUCH_MEAS_DONE |  |  |  | RTC_TOUCH_PAD_ACTIVE |  |  |  |       |  |   |
| 31         |   |   |   |   |   |   |   |   |   |   | 20 | 19                  | 16 | 15   | 14 |                     |  |  |  |                      |  |  |  |       |  | 0 |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0xf                 | 0  | 0x00 |    |                     |  |  |  |                      |  |  |  | Reset |  |   |

**RTC\_TOUCH\_PAD\_ACTIVE** Indicates whether the touch pin detects a touch. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

0: No touch detected

1: Touch detected

(RO)

**RTC\_TOUCH\_MEAS\_DONE** Indicates whether the touch pins have been measured at all frequency modes.

0: Not completed

1: Completed

(RO)

**RTC\_TOUCH\_SCAN\_CURR** Indicates the index number of the currently scanned touch pin. (RO)

Register 60.6. RTC\_TOUCH\_STATUS\_ *n* \_REG (*n*: 1-14) (0x0018+0x4\*(*n*-1))

|            |   |   |   |   |   |   |   |   |    |                                   |    |     |    |      |                                         |  |  |  |  |                                 |  |  |  |  |   |       |
|------------|---|---|---|---|---|---|---|---|----|-----------------------------------|----|-----|----|------|-----------------------------------------|--|--|--|--|---------------------------------|--|--|--|--|---|-------|
| (reserved) |   |   |   |   |   |   |   |   |    | RTC_TOUCH_PAD <sub>n</sub> NN_CNT |    |     |    |      | RTC_TOUCH_PAD <sub>n</sub> DEBOUNCE_CNT |  |  |  |  | RTC_TOUCH_PAD <sub>n</sub> DATA |  |  |  |  |   |       |
| 31         |   |   |   |   |   |   |   |   | 23 | 22                                | 19 | 18  | 16 | 15   |                                         |  |  |  |  |                                 |  |  |  |  | 0 |       |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0x0                               |    | 0x0 |    | 0x00 |                                         |  |  |  |  |                                 |  |  |  |  |   | Reset |

**RTC\_TOUCH\_PAD<sub>*n*</sub>DATA** Indicates the touch\_raw\_data, touch\_smooth\_data, or benchmark sampled by touch pin *n*. (RO)

**RTC\_TOUCH\_PAD<sub>*n*</sub>DEBOUNCE\_CNT** Indicates the cumulative number of consecutive touch or touch release detection by touch pin *n*. (RO)

**RTC\_TOUCH\_PAD<sub>*n*</sub>NN\_CNT** Indicates the cumulative number of consecutive instances where *touch\_smooth\_data* < *benchmark* – *passive\_noise\_threshold* has been detected by touch pin *n*. (RO)

Register 60.7. RTC\_TOUCH\_STATUS\_15\_REG (0x0050)

|                     |  |  |  |  |  |  |  |  |  |                      |  |     |  |                            |  |     |  |                    |  |    |  |  |  |  |  |       |  |   |  |
|---------------------|--|--|--|--|--|--|--|--|--|----------------------|--|-----|--|----------------------------|--|-----|--|--------------------|--|----|--|--|--|--|--|-------|--|---|--|
| (reserved)          |  |  |  |  |  |  |  |  |  | RTC_TOUCH_SLP_NN_CNT |  |     |  | RTC_TOUCH_SLP_DEBOUNCE_CNT |  |     |  | RTC_TOUCH_SLP_DATA |  |    |  |  |  |  |  |       |  |   |  |
| 31                  |  |  |  |  |  |  |  |  |  | 23                   |  | 22  |  | 19                         |  | 18  |  | 16                 |  | 15 |  |  |  |  |  |       |  | 0 |  |
| 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |  |  | 0x0                  |  | 0x0 |  | 0x0                        |  | 0x0 |  | 0x00               |  |    |  |  |  |  |  | Reset |  |   |  |

**RTC\_TOUCH\_SLP\_DATA** Indicates the touch\_smooth\_data or benchmark sampled by touch pin in sleep mode. (RO)

**RTC\_TOUCH\_SLP\_DEBOUNCE\_CNT** Indicates the cumulative number of consecutive touch or touch release detection by touch pin in sleep mode. (RO)

**RTC\_TOUCH\_SLP\_NN\_CNT** Indicates the cumulative number of consecutive instances where *touch\_smooth\_data* < *benchmark* – *passive\_noise\_threshold* has been detected by touch pin in sleep mode. (RO)

Register 60.8. RTC\_TOUCH\_STATUS\_16\_REG (0x0054)

|                            |  |  |  |  |  |  |  |     |  |  |  |  |  |  |  |                             |  |  |  |  |  |  |  |     |  |  |  |  |  |  |  |                             |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |                             |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |
|----------------------------|--|--|--|--|--|--|--|-----|--|--|--|--|--|--|--|-----------------------------|--|--|--|--|--|--|--|-----|--|--|--|--|--|--|--|-----------------------------|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|-----------------------------|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|
| RTC_TOUCH_SLP_APPROACH_CNT |  |  |  |  |  |  |  |     |  |  |  |  |  |  |  | RTC_TOUCH_APPROACH_PADO_CNT |  |  |  |  |  |  |  |     |  |  |  |  |  |  |  | RTC_TOUCH_APPROACH_PAD1_CNT |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  | RTC_TOUCH_APPROACH_PAD2_CNT |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |
| 31                         |  |  |  |  |  |  |  | 24  |  |  |  |  |  |  |  | 23                          |  |  |  |  |  |  |  | 16  |  |  |  |  |  |  |  | 15                          |  |  |  |  |  |  |  | 8 |  |  |  |  |  |  |  | 7                           |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |
| 0x0                        |  |  |  |  |  |  |  | 0x0 |  |  |  |  |  |  |  | 0x0                         |  |  |  |  |  |  |  | 0x0 |  |  |  |  |  |  |  | Reset                       |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |                             |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |

**RTC\_TOUCH\_APPROACH\_PAD2\_CNT** Indicates the cumulative number of samplings of touch pin 2 in proximity mode. (RO)

**RTC\_TOUCH\_APPROACH\_PAD1\_CNT** Indicates the cumulative number of samplings of touch pin 1 in proximity mode. (RO)

**RTC\_TOUCH\_APPROACH\_PADO\_CNT** Indicates the cumulative number of samplings of touch pin 0 in proximity mode. (RO)

**RTC\_TOUCH\_SLP\_APPROACH\_CNT** Indicates the cumulative number of samplings of the sleeping touch pin in proximity mode. (RO)

**Register 60.9. RTC\_TOUCH\_STATUS\_17\_REG (0x0058)**

|            |   |   |   |   |    |    |    |                   |   |  |  |                 |    |    |  |                  |   |  |    |                  |   |  |  |                    |   |   |   |                    |  |  |  |  |   |       |
|------------|---|---|---|---|----|----|----|-------------------|---|--|--|-----------------|----|----|--|------------------|---|--|----|------------------|---|--|--|--------------------|---|---|---|--------------------|--|--|--|--|---|-------|
| (reserved) |   |   |   |   |    |    |    | RTC_FREQ_SCAN_CNT |   |  |  | RTC_TOUCH_DBIAS |    |    |  | RTC_TOUCH_DRV_HS |   |  |    | RTC_TOUCH_DRV_LS |   |  |  | RTC_TOUCH_DRES_LPF |   |   |   | RTC_TOUCH_DCAP_LPF |  |  |  |  |   |       |
| 31         |   |   |   |   | 25 | 24 | 23 | 22                |   |  |  |                 | 18 | 17 |  |                  |   |  | 13 | 12               |   |  |  |                    | 9 | 8 | 7 | 6                  |  |  |  |  | 0 |       |
| 0          | 0 | 0 | 0 | 0 | 0  | 0  | 0  | 0                 | 0 |  |  |                 | 0  |    |  |                  | 0 |  |    |                  | 0 |  |  |                    | 0 | 0 |   |                    |  |  |  |  |   | Reset |

Reset

**RTC\_TOUCH\_DCAP\_LPF** Indicates the DCAP\_LPF value of the current touch pin. (RO)

**RTC\_TOUCH\_DRES\_LPF** Indicates the DRES\_LPF value of the current touch pin. (RO)

**RTC\_TOUCH\_DRV\_LS** Indicates the DRV\_LS value of the current touch pin. (RO)

**RTC\_TOUCH\_DRV\_HS** Indicates the DRV\_HS value of the current touch pin. (RO)

**RTC\_TOUCH\_DBIAS** Indicates the DBIAS value of the current touch pin. (RO)

**RTC\_FREQ\_SCAN\_CNT** Indicates the SCAN\_CNT value of the current touch pin. (RO)

**Register 60.10. RTC\_TOUCH\_CHN\_TMP\_STATUS\_REG (0x005C)**

|            |    |      |  |  |  |  |  |  |  |  |  |  |  |  |  |                             |    |    |  |  |  |  |  |  |  |  |  |  |  |       |  |                               |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------------|----|------|--|--|--|--|--|--|--|--|--|--|--|--|--|-----------------------------|----|----|--|--|--|--|--|--|--|--|--|--|--|-------|--|-------------------------------|---|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| (reserved) |    |      |  |  |  |  |  |  |  |  |  |  |  |  |  | RTC_TOUCH_PAD_ACTIVE_STATUS |    |    |  |  |  |  |  |  |  |  |  |  |  |       |  | RTC_TOUCH_PAD_INACTIVE_STATUS |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 31         | 30 | 29   |  |  |  |  |  |  |  |  |  |  |  |  |  |                             | 15 | 14 |  |  |  |  |  |  |  |  |  |  |  |       |  |                               | 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0          | 0  | 0x00 |  |  |  |  |  |  |  |  |  |  |  |  |  | 0x00                        |    |    |  |  |  |  |  |  |  |  |  |  |  | Reset |  |                               |   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

Reset

**RTC\_TOUCH\_PAD\_INACTIVE\_STATUS** Indicates whether the touch pin detects the touch pad from being touched to being released. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

0: Touch release detected

1: Touch release not detected

(RO)

**RTC\_TOUCH\_PAD\_ACTIVE\_STATUS** Indicates whether the touch pin detects the process of being touched. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

0: Touch detected

1: Touch not detected

(RO)

Register 60.11. RTC\_TOUCH\_DATE\_REG (0x0100)

|                          |    |    |    |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |
|--------------------------|----|----|----|-----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------|
| RTC_CLK_EN<br>(reserved) |    |    |    | RTC_DATE  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0     |
| 31                       | 30 | 28 | 27 |           |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |
| 0                        | 0  | 0  | 0  | 0x2309130 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Reset |

**RTC\_CLK\_EN** Configures whether to enable the register read/write clock.

0: Disable

1: Enable

(R/W)

**RTC\_DATE** Version control register. (R/W)

## 60.7.2 Configuration Registers

The addresses in this section are relative to LP Analog Peripheral base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 60.12. LP\_ANA\_TOUCH\_APPROACH\_WORK\_MEAS\_NUM\_REG (0x00FC)

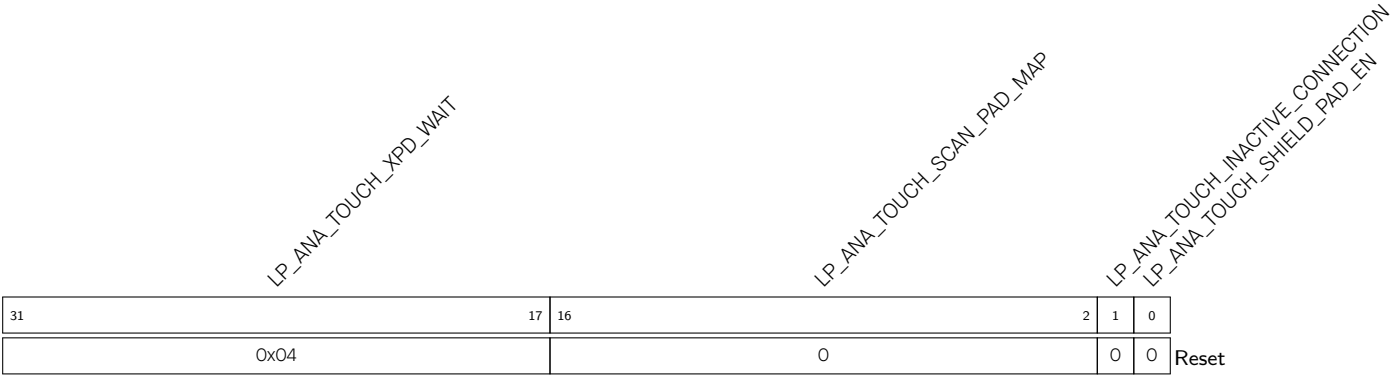
|            |    |     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     |    |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     |  |  |    |   |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |   |  |  |  |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------------|----|-----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----|----|---------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-----|--|--|----|---|--|---------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|-------|--|--|--|--|---|--|--|--|--|---------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| (reserved) |    |     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     |    | LP_ANA_TOUCH_APPROACH_MEAS_NUM0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     |  |  |    |   |  | LP_ANA_TOUCH_APPROACH_MEAS_NUM1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |   |  |  |  |  | LP_ANA_TOUCH_APPROACH_MEAS_NUM2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 31         | 30 | 29  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     | 20 | 19                              |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |     |  |  | 10 | 9 |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  | 0 |  |  |  |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0          | 0  | 100 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 100 |    |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 100 |  |  |    |   |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | Reset |  |  |  |  |   |  |  |  |  |                                 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

**LP\_ANA\_TOUCH\_APPROACH\_MEAS\_NUM2** Configures the number of measurements for sampling frequency mode 2 in proximity mode. (R/W)

**LP\_ANA\_TOUCH\_APPROACH\_MEAS\_NUM1** Configures the number of measurements for sampling frequency mode 1 in proximity mode. (R/W)

**LP\_ANA\_TOUCH\_APPROACH\_MEAS\_NUM0** Configures the number of measurements for sampling frequency mode 0 in proximity mode. (R/W)

Register 60.13. LP\_ANA\_TOUCH\_SCAN\_CTRL1\_REG (0x0100)



**LP\_ANA\_TOUCH\_SHIELD\_PAD\_EN** Configures whether to enable the moisture tolerance feature.  
0: Disable  
1: Enable  
(R/W)

**LP\_ANA\_TOUCH\_INACTIVE\_CONNECTION** Configures whether to power up the touch pins involved in the measurement in scan mode.  
0: Not power up  
1: Power up  
(R/W)

**LP\_ANA\_TOUCH\_SCAN\_PAD\_MAP** Configures which touch pins to use in scan mode for measurement. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.  
0: No effect  
1: Enable the touch pin  
(R/W)

**LP\_ANA\_TOUCH\_XPD\_WAIT** Configures the power-up wait time after initiating a measurement.  
(R/W)

**Register 60.14. LP\_ANA\_TOUCH\_SCAN\_CTRL2\_REG (0x0104)**

|            |    |                            |    |                     |     |                       |    |                         |  |                          |   |  |  |  |  |            |   |
|------------|----|----------------------------|----|---------------------|-----|-----------------------|----|-------------------------|--|--------------------------|---|--|--|--|--|------------|---|
| (reserved) |    | LP_ANA_FREQ_SCAN_CNT_LIMIT |    | LP_ANA_FREQ_SCAN_EN |     | LP_ANA_TOUCH_OUT_RING |    | LP_ANA_TOUCH_TIMEOUT_EN |  | LP_ANA_TOUCH_TIMEOUT_NUM |   |  |  |  |  | (reserved) |   |
| 31         | 30 | 29                         | 28 | 27                  | 26  | 23                    | 22 | 21                      |  | 6                        | 5 |  |  |  |  |            | 0 |
| 0          | 0  |                            | 3  | 0                   | 0xf |                       | 0  |                         |  | 0xffff                   |   |  |  |  |  | 0          | 0 |

Reset

**LP\_ANA\_TOUCH\_TIMEOUT\_NUM** Configures the number of measurement timeouts. (R/W)

**LP\_ANA\_TOUCH\_TIMEOUT\_EN** Configures whether to enable the measurement timeout.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_TOUCH\_OUT\_RING** Configures which touch pins to be used for water rejection. Configurable values are 1-14. Other values are invalid. (R/W)

**LP\_ANA\_FREQ\_SCAN\_EN** Configures whether to enable frequency hopping.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_FREQ\_SCAN\_CNT\_LIMIT** Configures the number of frequency hops.

0: 1

1: 2

2: 3

3: Invalid

(R/W)

**Register 60.15. LP\_ANA\_TOUCH\_WORK\_REG (0x0108)**

|            |   |   |   |                       |    |            |    |                      |  |                 |    |                 |  |                 |    |            |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
|------------|---|---|---|-----------------------|----|------------|----|----------------------|--|-----------------|----|-----------------|--|-----------------|----|------------|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|
| (reserved) |   |   |   | LP_ANA_TOUCH_OUT_GATE |    | (reserved) |    | LP_ANA_TOUCH_OUT_SEL |  | LP_ANA_DIV_NUM0 |    | LP_ANA_DIV_NUM1 |  | LP_ANA_DIV_NUM2 |    | (reserved) |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| 31         |   |   |   | 28                    | 27 | 26         | 25 | 24                   |  |                 | 22 | 21              |  |                 | 19 | 18         |   |   | 16 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |
| 0          | 0 | 0 | 0 | 0                     | 0  | 0          | 0  | 0x0                  |  | 0x0             |    | 0x0             |  | 0               |    | 0          | 0 | 0 | 0  | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0</ |

**LP\_ANA\_DIV\_NUM2** Configures the enabling of the TOUCH\_OUT signal division and the coefficient for sampling frequency mode 0.

bit[0]: Configures whether to enable division.

0: Disable

1: Enable

bit[1-2]: Configures the TOUCH\_OUT signal division coefficient.

0: 0

1: 2

2: 4

3: 6

(R/W)

**LP\_ANA\_DIV\_NUM1** Configures the enabling of the TOUCH\_OUT signal division and the coefficient for sampling frequency mode 1. For detailed configuration, refer to [LP\\_ANA\\_DIV\\_NUM2](#). (R/W)

**LP\_ANA\_DIV\_NUM0** Configures the enabling of the TOUCH\_OUT signal division and the coefficient for sampling frequency mode 2. For detailed configuration, refer to [LP\\_ANA\\_DIV\\_NUM2](#). (R/W)

**LP\_ANA\_TOUCH\_OUT\_SEL** Configures whether to use the TOUCH\_OUT signal for clock or data.

0: TOUCH\_OUT signal is used as data

1: TOUCH\_OUT signal is used as clock

(R/W)

**LP\_ANA\_TOUCH\_OUT\_GATE** Configures whether to enable gating after enabling the TOUCH\_OUT signal division.

0: Disable

1: Enable

(R/W)

Register 60.16. LP\_ANA\_TOUCH\_WORK\_MEAS\_NUM\_REG (0x010C)

|            |    |                        |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                        |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                        |  |  |    |   |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |   |
|------------|----|------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------------------------|----|----|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|------------------------|--|--|----|---|--|--|--|--|--|--|--|--|--|--|--|--|--|-------|--|--|--|--|---|
| (reserved) |    | LP_ANA_TOUCH_MEAS_NUM0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_MEAS_NUM1 |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_MEAS_NUM2 |  |  |    |   |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  |   |
| 31         | 30 | 29                     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                        | 20 | 19 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                        |  |  | 10 | 9 |  |  |  |  |  |  |  |  |  |  |  |  |  |       |  |  |  |  | 0 |
| 0          | 0  | 100                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 100                    |    |    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 100                    |  |  |    |   |  |  |  |  |  |  |  |  |  |  |  |  |  | Reset |  |  |  |  |   |

- LP\_ANA\_TOUCH\_MEAS\_NUM2 Configures the number of measurements for sampling frequency mode 2 in normal operation. (R/W)
- LP\_ANA\_TOUCH\_MEAS\_NUM1 Configures the number of measurements for sampling frequency mode 1 in normal operation. (R/W)
- LP\_ANA\_TOUCH\_MEAS\_NUM0 Configures the number of measurements for sampling frequency mode 0 in normal operation.(R/W)

**Register 60.17. LP\_ANA\_TOUCH\_FILTER1\_REG (0x0110)**

|                             |    |    |                             |    |    |                       |    |    |                        |   |   |                          |   |   |                          |   |   |                         |   |   |                          |  |  |                       |  |  |                         |  |  |                                     |  |  |
|-----------------------------|----|----|-----------------------------|----|----|-----------------------|----|----|------------------------|---|---|--------------------------|---|---|--------------------------|---|---|-------------------------|---|---|--------------------------|--|--|-----------------------|--|--|-------------------------|--|--|-------------------------------------|--|--|
| LP_ANA_TOUCH_DEBOUNCE_LIMIT |    |    | LP_ANA_TOUCH_APPROACH_LIMIT |    |    | LP_ANA_TOUCH_NN_LIMIT |    |    | LP_ANA_TOUCH_FILTER_EN |   |   | LP_ANA_TOUCH_FILTER_MODE |   |   | LP_ANA_TOUCH_JITTER_STEP |   |   | LP_ANA_TOUCH_SMOOTH_LVL |   |   | LP_ANA_TOUCH_NOISE_THRES |  |  | LP_ANA_TOUCH_NN_THRES |  |  | LP_ANA_TOUCH_HYSTERESIS |  |  | LP_ANA_TOUCH_NN_DISUPDATE_BENCHMARK |  |  |
| 31                          | 29 | 28 | 21                          | 20 | 17 | 16                    | 15 | 13 | 12                     | 9 | 8 | 7                        | 6 | 5 | 4                        | 3 | 2 | 1                       | 0 | 0 | Reset                    |  |  |                       |  |  |                         |  |  |                                     |  |  |
| 3                           |    | 80 |                             |    | 5  |                       | 0  | 0  |                        | 1 |   | 0                        | 0 | 0 | 0                        | 0 | 0 | 0                       | 0 |   |                          |  |  |                       |  |  |                         |  |  |                                     |  |  |

**LP\_ANA\_TOUCH\_NN\_DISUPDATE\_BENCHMARK\_EN** Configures whether to turn off the benchmark data correction.

0: Do not turn off

1: Turn off

(R/W)

**LP\_ANA\_TOUCH\_HYSTERESIS** Configures the sampling hysteresis. (R/W)

**LP\_ANA\_TOUCH\_NN\_THRES** Configures the out-of-phase noise\_threshold. (R/W)

**LP\_ANA\_TOUCH\_NOISE\_THRES** Configures the noise\_threshold. (R/W)

**LP\_ANA\_TOUCH\_SMOOTH\_LVL** Configures the IIR filter module for touch\_smooth\_data.(R/W)

**LP\_ANA\_TOUCH\_JITTER\_STEP** Configures the jitter step in filter module 7 for generating benchmark data. (R/W)

**LP\_ANA\_TOUCH\_FILTER\_MODE** Configures the filter module for generating benchmark data. (R/W)

**LP\_ANA\_TOUCH\_FILTER\_EN** Configures whether to enable the filter module for generating benchmark data.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_TOUCH\_NN\_LIMIT** Configures the number of consecutive sampling instances where the sampling value exceeds the out-of-phase noise\_threshold. (R/W)

**LP\_ANA\_TOUCH\_APPROACH\_LIMIT** Configures the number of sampling in proximity mode. (R/W)

**LP\_ANA\_TOUCH\_DEBOUNCE\_LIMIT** Configures the number of consecutive sampling for detecting touch state changes. (R/W)

### Register 60.18. LP\_ANA\_TOUCH\_FILTER2\_REG (0x0114)

|    |    |        |    |    |                                 |
|----|----|--------|----|----|---------------------------------|
| 31 | 30 | 29     | 15 | 14 | 0                               |
| 0  | 0  | 0x3fff |    |    | 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |

LP\_ANA\_TOUCH\_BYPASS\_NN\_THRES  
LP\_ANA\_TOUCH\_BYPASS\_NOISE\_THRES  
LP\_ANA\_TOUCH\_OUTEN  
(reserved)  
Reset

**LP\_ANA\_TOUCH\_OUTEN** Configures whether to enable output of the 14 touch pin measurement results. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_TOUCH\_BYPASS\_NOISE\_THRES** Configures whether to enable the bypass mode for touch\_smooth\_data.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_TOUCH\_BYPASS\_NN\_THRES** Configures whether to enable the bypass mode for out-of-phase noise threshold.

0: Disable

1: Enable

(R/W)

### Register 60.19. LP\_ANA\_TOUCH\_FILTER3\_REG (0x0118)

|                                 |                                         |                                  |
|---------------------------------|-----------------------------------------|----------------------------------|
| (reserved)                      |                                         |                                  |
|                                 | <i>LP_ANA_TOUCH_UPDATE_BENCHMARK_SW</i> | <i>LP_ANA_TOUCH_BENCHMARK_SW</i> |
| 31                              | 17                                      | 0                                |
| 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 | 0                                       | 0                                |
|                                 |                                         |                                  |
| Reset                           |                                         |                                  |

**LP\_ANA\_TOUCH\_BENCHMARK\_SW** Configures software to write the benchmark data. (R/W)

LP\_ANA\_TOUCH\_UPDATE\_BENCHMARK\_SW Write 1 to enable the software to write the benchmark data. (WT)

### Register 60.20. LP\_ANA\_TOUCH\_SLPO\_REG (0x011C)

|                         |  |  |  |  |  |  |  |  |  |  |  |                      |  |  |  |                              |  |  |  |                      |  |  |  |    |  |  |  |    |  |  |  |       |  |  |  |  |  |  |  |   |  |  |  |
|-------------------------|--|--|--|--|--|--|--|--|--|--|--|----------------------|--|--|--|------------------------------|--|--|--|----------------------|--|--|--|----|--|--|--|----|--|--|--|-------|--|--|--|--|--|--|--|---|--|--|--|
| (reserved)              |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_SLP_PAD |  |  |  | LP_ANA_TOUCH_SLP_CHANNEL_CLR |  |  |  | LP_ANA_TOUCH_SLP_TH0 |  |  |  |    |  |  |  |    |  |  |  |       |  |  |  |  |  |  |  |   |  |  |  |
| 31                      |  |  |  |  |  |  |  |  |  |  |  | 21                   |  |  |  | 20                           |  |  |  | 17                   |  |  |  | 16 |  |  |  | 15 |  |  |  |       |  |  |  |  |  |  |  | 0 |  |  |  |
| 0 0 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |  |  |  |  | 0xf                  |  |  |  | 0                            |  |  |  | 0                    |  |  |  |    |  |  |  |    |  |  |  | Reset |  |  |  |  |  |  |  |   |  |  |  |

**LP\_ANA\_TOUCH\_SLP\_TH0** Configures the touch threshold for sampling frequency mode 0 in sleep mode. (R/W)

**LP\_ANA\_TOUCH\_SLP\_CHANNEL\_CLR** Write 1 to clear the benchmark data in sleep mode. (WT)

**LP\_ANA\_TOUCH\_SLP\_PAD** Configures which touch sensor to enter sleep mode. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

0: No effect

1: Selects the touch sensor to enter sleep mode

(R/W)

### Register 60.21. LP\_ANA\_TOUCH\_SLP1\_REG (0x0120)

|                      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| LP_ANA_TOUCH_SLP_TH1 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_SLP_TH2 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 31                   |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 0                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0                    |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| Reset                |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |                      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

**LP\_ANA\_TOUCH\_SLP\_TH2** Configures the touch threshold for sampling frequency mode 2 in sleep mode. (R/W)

**LP\_ANA\_TOUCH\_SLP\_TH1** Configures the touch threshold for sampling frequency mode 1 in sleep mode. (R/W)

**Register 60.22. LP\_ANA\_TOUCH\_CLR\_REG (0x0124)**

|            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |                         |    |   |  |  |  |  |  |                          |  |       |  |  |  |  |  |
|------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|-------------------------|----|---|--|--|--|--|--|--------------------------|--|-------|--|--|--|--|--|
| (reserved) |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    | LP_ANA_TOUCH_STATUS_CLR |    |   |  |  |  |  |  | LP_ANA_TOUCH_CHANNEL_CLR |  |       |  |  |  |  |  |
| 31         |   |   |   |   |   |   |   |   |   |   |   |   |   |   | 16 | 15                      | 14 |   |  |  |  |  |  |                          |  | 0     |  |  |  |  |  |
| 0          | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0  | 0                       | 0  | 0 |  |  |  |  |  |                          |  | Reset |  |  |  |  |  |

**LP\_ANA\_TOUCH\_CHANNEL\_CLR** Write 1 to independently clear the touch state of the 14 touch sensors. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid. (WT)

**LP\_ANA\_TOUCH\_STATUS\_CLR** Write 1 to clear the touch state. (WT)

**Register 60.23. LP\_ANA\_TOUCH\_APPROACH\_REG (0x0128)**

|                                         |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |     |  |  |                              |     |  |  |  |     |  |  |                            |       |   |  |  |  |   |   |                            |  |  |  |  |  |  |  |                            |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----|-----|--|--|------------------------------|-----|--|--|--|-----|--|--|----------------------------|-------|---|--|--|--|---|---|----------------------------|--|--|--|--|--|--|--|----------------------------|--|--|--|--|--|--|--|
| (reserved)                              |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |     |  |  | LP_ANA_TOUCH_SLP_APPROACH_EN |     |  |  |  |     |  |  | LP_ANA_TOUCH_APPROACH_PAD2 |       |   |  |  |  |   |   | LP_ANA_TOUCH_APPROACH_PAD1 |  |  |  |  |  |  |  | LP_ANA_TOUCH_APPROACH_PAD0 |  |  |  |  |  |  |  |
| 31                                      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13 |     |  |  | 12                           | 11  |  |  |  | 8   |  |  |                            | 7     | 4 |  |  |  | 3 | 0 |                            |  |  |  |  |  |  |  |                            |  |  |  |  |  |  |  |
| 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 0  | 0xf |  |  |                              | 0xf |  |  |  | 0xf |  |  |                            | Reset |   |  |  |  |   |   |                            |  |  |  |  |  |  |  |                            |  |  |  |  |  |  |  |

**LP\_ANA\_TOUCH\_APPROACH\_PAD0** Configure the first touch pin used in proximity mode. Configurable values are 1-14. Other values are invalid. (R/W)

**LP\_ANA\_TOUCH\_APPROACH\_PAD1** Configure the second touch pin used in proximity mode. Configurable values are 1-14. Other values are invalid. (R/W)

**LP\_ANA\_TOUCH\_APPROACH\_PAD2** Configure the third touch pin used in proximity mode. Configurable values are 1-14. Other values are invalid. (R/W)

**LP\_ANA\_TOUCH\_SLP\_APPROACH\_EN** Configures whether to enable proximity mode.

0: Disable

1: Enable

(R/W)

**Register 60.24. LP\_ANA\_TOUCH\_FREQ $n$ \_SCAN\_PARA\_REG ( $n$ : 0-2) (0x012C+0x4\* $n$ )**

|            |  |  |  |  |  |  |  |  |  |                                       |  |  |  |  |  |  |  |                                        |  |    |  |  |  |  |  |                                        |  |  |  |    |  |  |  |                                          |  |  |  |  |  |    |  |                                          |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |
|------------|--|--|--|--|--|--|--|--|--|---------------------------------------|--|--|--|--|--|--|--|----------------------------------------|--|----|--|--|--|--|--|----------------------------------------|--|--|--|----|--|--|--|------------------------------------------|--|--|--|--|--|----|--|------------------------------------------|--|--|--|--|--|--|--|----|--|--|--|--|--|--|--|--|--|----|--|--|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|--|--|---|--|--|--|--|--|--|--|--|--|-------|--|--|--|--|--|--|--|--|--|
| (reserved) |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_FREQ <sub>n</sub> _DBIAS |  |  |  |  |  |  |  | LP_ANA_TOUCH_FREQ <sub>n</sub> _DRV_HS |  |    |  |  |  |  |  | LP_ANA_TOUCH_FREQ <sub>n</sub> _DRV_LS |  |  |  |    |  |  |  | LP_ANA_TOUCH_FREQ <sub>n</sub> _DRES_LPF |  |  |  |  |  |    |  | LP_ANA_TOUCH_FREQ <sub>n</sub> _DCAP_LPF |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |  |  |    |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |   |  |  |  |  |  |  |  |  |  |       |  |  |  |  |  |  |  |  |  |
| 31         |  |  |  |  |  |  |  |  |  | 23                                    |  |  |  |  |  |  |  |                                        |  | 22 |  |  |  |  |  |                                        |  |  |  | 18 |  |  |  |                                          |  |  |  |  |  | 17 |  |                                          |  |  |  |  |  |  |  | 13 |  |  |  |  |  |  |  |  |  | 12 |  |  |  |  |  |  |  |  |  | 9 |  |  |  |  |  |  |  |  |  | 8 |  |  |  |  |  |  |  |  |  | 7 |  |  |  |  |  |  |  |  |  | 6 |  |  |  |  |  |  |  |  |  | 0     |  |  |  |  |  |  |  |  |  |
| 0          |  |  |  |  |  |  |  |  |  | 0                                     |  |  |  |  |  |  |  |                                        |  | 0  |  |  |  |  |  |                                        |  |  |  | 0  |  |  |  |                                          |  |  |  |  |  | 0  |  |                                          |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  |  |  | 0  |  |  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |  |  | 0 |  |  |  |  |  |  |  |  |  | Reset |  |  |  |  |  |  |  |  |  |

**LP\_ANA\_TOUCH\_FREQ $n$ \_DCAP\_LPF** Configures the touch sensor low-pass filter for sampling frequency mode  $n$ . The adjustment range is 0-2.54 pF, and the step is 20 fF. (R/W)(R/W)

**LP\_ANA\_TOUCH\_FREQ $n$ \_DRES\_LPF** Configures the touch sensor low-pass filter for sampling frequency mode  $n$ .

0: 0 k $\Omega$

1: 3 k $\Omega$

2: 4.5 k $\Omega$

3: 7.5 k $\Omega$

(R/W)

**LP\_ANA\_TOUCH\_FREQ $n$ \_DRV\_LS** Configures the value of DRV\_LS of the touch sensor for sampling frequency mode  $n$ . (R/W)

**LP\_ANA\_TOUCH\_FREQ $n$ \_DRV\_HS** Configures the value of DRV\_HS of the touch sensor for sampling frequency mode  $n$ . (R/W)

**LP\_ANA\_TOUCH\_FREQ $n$ \_DBIAS** Configures whether to enable the moisture tolerance function and configures the internal voltage of the touch sensor for sampling frequency mode  $n$ .

bit[0]: Configures whether to enable the moisture tolerance function for sampling frequency mode  $n$ .

0: Disable

1: Enable

bit[1-4]: Configures the internal voltage of the touch sensor for sampling frequency mode  $n$ .

(R/W)

### Register 60.25. LP\_ANA\_TOUCH\_ANA\_PARA\_REG (0x0138)

|                     |  |  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |  |  |                             |  |  |  |                           |  |   |  |                            |  |   |  |   |  |
|---------------------|--|--|--|--|--|--|--|--|--|---------------------|--|--|--|--|--|--|--|--|--|-----------------------------|--|--|--|---------------------------|--|---|--|----------------------------|--|---|--|---|--|
| (reserved)          |  |  |  |  |  |  |  |  |  |                     |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_TOUCH_DCAP_CAL |  |  |  | LP_ANA_TOUCH_TOUCH_EN_CAL |  |   |  | LP_ANA_TOUCH_TOUCH_BUF_DRV |  |   |  |   |  |
| 31                  |  |  |  |  |  |  |  |  |  | 11                  |  |  |  |  |  |  |  |  |  | 10                          |  |  |  | 4                         |  |   |  | 3                          |  | 2 |  | 0 |  |
| 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |  |  | 0 0 0 0 0 0 0 0 0 0 |  |  |  |  |  |  |  |  |  | 0                           |  |  |  | 0                         |  | 0 |  | Reset                      |  |   |  |   |  |

**LP\_ANA\_TOUCH\_TOUCH\_BUF\_DRV** Configures the value of BUF\_DRV. (R/W)

**LP\_ANA\_TOUCH\_TOUCH\_EN\_CAL** Configures whether to enable the touch sensor internal capacitance test mode.

0: Disable

1: Enable

(R/W)

**LP\_ANA\_TOUCH\_TOUCH\_DCAP\_CAL** Configures the internal capacitance connected to the touch pins. The default value is 0, and the adjustment range is 0-127 pF with a step of 1. (R/W)

**Register 60.26. LP\_ANA\_TOUCH\_MUX0\_REG (0x013C)**

|                          |    |    |    |    |                     |    |    |    |   |   |   |                       |   |   |   |                       |   |   |   |            |   |   |   |
|--------------------------|----|----|----|----|---------------------|----|----|----|---|---|---|-----------------------|---|---|---|-----------------------|---|---|---|------------|---|---|---|
| LP_ANA_TOUCH_START_FORCE |    |    |    |    | LP_ANA_TOUCH_BUFSEL |    |    |    |   |   |   | LP_ANA_TOUCH_FREQ_SEL |   |   |   | LP_ANA_TOUCH_DATA_SEL |   |   |   | (reserved) |   |   |   |
| 31                       | 30 | 29 | 28 | 27 | 26                  | 12 | 11 | 10 | 9 | 8 | 7 |                       |   |   |   |                       |   |   |   |            |   |   |   |
| 0                        | 0  | 1  | 0  | 0  |                     | 0  |    |    | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0                     | 0 | 0 | 0 | 0          | 0 | 0 | 0 |

Reset

**LP\_ANA\_TOUCH\_DATA\_SEL** Configures the type of the return value.

- 0/1: touch\_raw\_data
  - 2: benchmark
  - 3: touch\_smooth\_data
- (R/W)

**LP\_ANA\_TOUCH\_FREQ\_SEL** Configures the corresponding sampling frequency mode of the return value.

- 0: 0
  - 1: 1
  - 2: 2
  - 3: Invalid
- (R/W)

**LP\_ANA\_TOUCH\_BUFSEL** Configures whether to enable a touch pin to be used for moisture tolerance. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.

- 0: Disable
  - 1: Enable
- (R/W)

**LP\_ANA\_TOUCH\_DONE\_EN** Configures whether to enable software to trigger the DONE signal that enables the measurement.

- 0: Disable
  - 1: Enable
- (R/W)

**LP\_ANA\_TOUCH\_DONE\_FORCE** Configures whether to generate the DONE signal to end the current measurement operation.

- 0: Not generate
  - 1: Generate
- (R/W)

**LP\_ANA\_TOUCH\_FSM\_EN** Configures the control source to power up and activate touch pins.

- 0: Controlled by software
  - 1: Controlled by hardware
- (R/W)

Continued on the next page...

Register 60.26. LP\_ANA\_TOUCH\_MUX0\_REG (0x013C)

Continued from the previous page...

**LP\_ANA\_TOUCH\_START\_EN** Configures whether to enable software to trigger the START signal to start the measurement.  
0: Disable  
1: Enable  
(R/W)

**LP\_ANA\_TOUCH\_START\_FORCE** Configures whether to generate the DONE signal to start a measurement.  
0: Not generate  
1: Generate  
(R/W)

Register 60.27. LP\_ANA\_TOUCH\_MUX1\_REG (0x0140)

|            |    |                  |  |  |  |  |  |  |  |  |  |  |  |  |  |                    |    |  |  |  |  |  |  |  |  |  |  |  |  |       |   |
|------------|----|------------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--------------------|----|--|--|--|--|--|--|--|--|--|--|--|--|-------|---|
| (reserved) |    | LP_ANA_TOUCH_XPD |  |  |  |  |  |  |  |  |  |  |  |  |  | LP_ANA_TOUCH_START |    |  |  |  |  |  |  |  |  |  |  |  |  |       |   |
| 31         | 30 | 29               |  |  |  |  |  |  |  |  |  |  |  |  |  | 15                 | 14 |  |  |  |  |  |  |  |  |  |  |  |  |       | 0 |
| 0          | 0  | 0                |  |  |  |  |  |  |  |  |  |  |  |  |  | 0                  |    |  |  |  |  |  |  |  |  |  |  |  |  | Reset |   |

**LP\_ANA\_TOUCH\_START** Configures whether to activate the 14 touch pins. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.  
0: No effect  
1: Activate  
(R/W)

**LP\_ANA\_TOUCH\_XPD** Configures whether to power up the 14 touch pins. Bit 1-14 corresponds to touch pin 1-14 respectively. Other bits are invalid.  
0: Not power up  
1: Power up  
(R/W)

LP\_ANA\_TOUCH\_PADn\_TH0

(reserved)

Register 60.29. LP\_ANA\_TOUCH\_PAD $n$ \_TH1\_REG ( $n$ : 1-14) (0x0154+0xC\*( $n$ -1))

LP\_ANA\_TOUCH\_PAD<sup>n</sup>\_TH1

Register 60.30. LP\_ANA\_TOUCH\_PAD $n$ \_TH2\_REG ( $n$ : 1-14) (0x0158+0xC\*( $n$ -1))

LP\_ANA\_TOUCH\_PADn\_TH2

ESP32-P4 TRM  
PRELIMINARY

### Register 60.31. LP\_ANA\_TOUCH\_CTRL\_REG (0x01FC)

[illegible]

**LP\_ANA\_TOUCH\_UPDATE\_BENCHMARK\_FREQ\_SEL** Configures the frequency point used by software to update the benchmark. (R/W)

**LP\_ANA\_TOUCH\_UPDATE\_BENCHMARK\_PAD\_SEL** Configures the channel used by software to update the benchmark. (R/W)

**LP\_ANA\_FREQ\_SCAN\_CNT\_RISE** Configures the number of hit frequency points to be evaluated for touch detection in frequency hopping mode. (R/W)

### Register 60.32. LP\_ANA\_DATE\_REG (0x03FC)

Diagram of the LP\_ANA\_LP\_ANA\_DATE register. The register is 32 bits wide. Bit 31 is labeled LP\_ANA\_CLK\_EN. Bits 30 down to 1 are labeled LP\_ANA\_LP\_ANA\_DATE. The register value is 0x250220.

**LP\_ANA\_LP\_ANA\_DATE** Version control register. (R/W)

**LP\_ANA\_CLK\_EN** Configures whether to enable the clock for accessing the registers.

0: Disable

1: Enable

(R/W)